

Mathematics of Machine Learning

Bayesian Inference



Bayes, Conjugate Priors, Uninformative Priors

Outline

1. **Bayes' Rule**
2. Conjugate Priors
3. Uninformative Priors



Dealing with Uncertainty

- In machine learning we are attempting to make inference under uncertainty
- The natural language for discussing uncertainty is probability
- The natural framework for making inferences is Bayesian statistics
- However, this requires that we encode our prior knowledge of the problem and specify a likelihood
- In consequence, probabilistic methods tend to be bespoke, rather than general purpose black boxes

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Revision on Bayes

- Bayes' rule

$$\mathbb{P}(\mathcal{H}_i|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\mathcal{H}_i)\mathbb{P}(\mathcal{H}_i)}{\mathbb{P}(\mathcal{D})}$$

- ★ $\mathbb{P}(\mathcal{H}_i|\mathcal{D})$ is the **posterior** probability of a hypothesis $\mathcal{H}_i$ (i.e. the probability of $\mathcal{H}_i$ **after** we see the data)
- ★ $\mathbb{P}(\mathcal{D}|\mathcal{H}_i)$ is the **likelihood** of the data given the hypothesis.
Note, that we calculated this from the forward problem
- ★ $\mathbb{P}(\mathcal{H}_i)$ is the **prior** probability (i.e. the probability of $\mathcal{H}_i$ **before** we see the data)
- ★ $\mathbb{P}(\mathcal{D})$ is the **evidence**

$$\mathbb{P}(\mathcal{D}) = \sum_{i=1}^n \mathbb{P}(\mathcal{H}_i, \mathcal{D}) = \sum_{i=1}^n \mathbb{P}(\mathcal{D}|\mathcal{H}_i)\mathbb{P}(\mathcal{H}_i)$$

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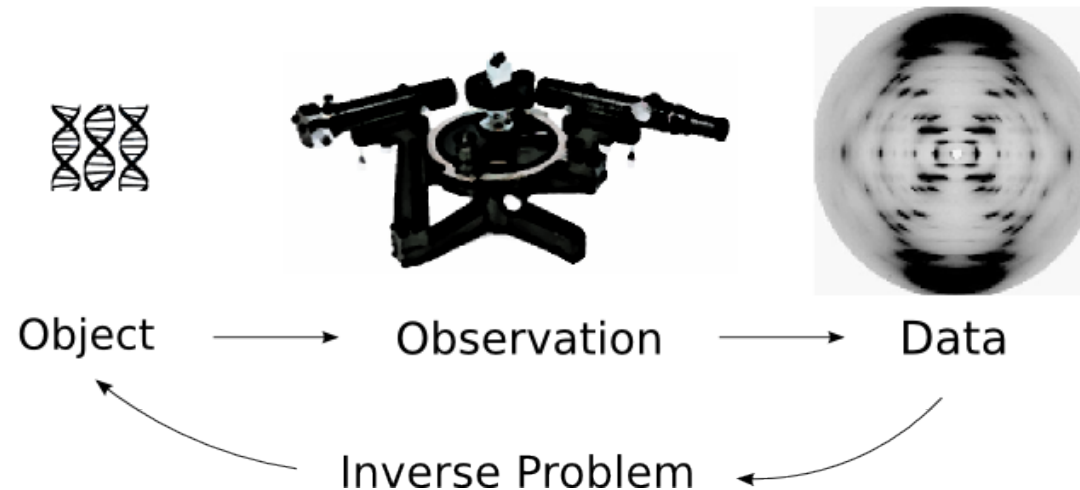
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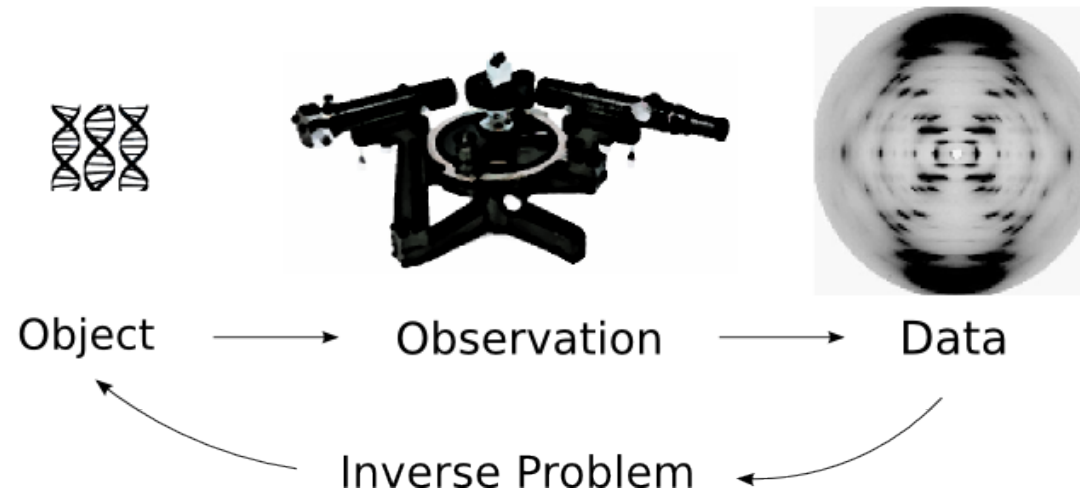
Solving Inverse Problems



- We want the posterior $\mathbb{P}(\mathcal{H}_i|\mathcal{D})$ (i.e. the probability of what happened given some evidence)
- The Bayesian formalism converts this into the forward problem

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- We calculate the likelihood $\mathbb{P}(\mathcal{D}|\mathcal{H}_i)$ (i.e. assuming the hypothesis, what is the chance of obtaining the data?)
- We consider the process of how the data is generated
- This uses the data we have (doesn't care about missing data)
- But we also need to know the prior $\mathbb{P}(\mathcal{H}_i)$
- Also, this can get difficult when we have many hypotheses

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Evidence

- The normalisation term

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tells you how likely the data is (given the prior and likelihood function)

- It is called the **marginal likelihood**
- At one level this is just a normalisation condition
- But it also plays an important role in **model selection**

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Model Selection

- Suppose we have two models of the likelihood or prior; M_1 and M_2 . Then the posterior will depend on the model

$$\mathbb{P}(\mathcal{H} \mid \mathcal{D}, M_i) = \frac{\mathbb{P}(\mathcal{D} \mid \mathcal{H}, M_i) \mathbb{P}(\mathcal{H} \mid M_i)}{\mathbb{P}(\mathcal{D} \mid M_i)}$$

- We can use $\mathbb{P}(\mathcal{D} \mid M_1)$ or $\mathbb{P}(\mathcal{D} \mid M_2)$ to select which model is most likely!
- E.g. is my likelihood best captured by a normal or gamma distribution
- This also allows us to select hyperparameters for a model

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Probability Density

- When we are working with continuous variables it is more natural to work with probability densities

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x \leq X < x + \delta x)}{\delta x}$$

- Note that densities are non-negative, but can be greater than 1 (they are not probabilities)
- However

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

is a probability and is less than or equal to 1

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Densities and Bayes

- Bayes' rule also applies to densities

$$\mathbb{P}(x \leq X < x + \delta x | Y) = \frac{\mathbb{P}(Y|x) \mathbb{P}(x \leq X < x + \delta x)}{\mathbb{P}(Y)}$$

- Dividing by δx and taking the limit $\delta x \rightarrow 0$

$$f_{X|Y}(x|Y) = \frac{\mathbb{P}(Y|x) f_X(x)}{\mathbb{P}(Y)}$$

- Similarly if X is discrete and Y continuous

$$\mathbb{P}(X|y) = \frac{f_{Y|X}(y|X) \mathbb{P}(X)}{f_Y(y)}$$

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Practical Bayesian Inference

- Often consider learning parameters θ

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

- This can be hard for large data sets as the posterior, $p(\theta|\mathcal{D})$, is often a mess
- If we are lucky and have a simple likelihood then if we choose the right prior we end up with a posterior of the same form as the prior
- This occurs in some classic probabilistic inference problems, but as we will see soon it is also true for Gaussian Processes

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Learning a Probability

- Suppose we have a coin and we want to establish the probability of a head
- We want to learn this from a series of independent trials
- (Independent trials with two possible outcomes are known in probability theory as Bernoulli trials)
- Let X_i equal 1 if the i^{th} trial is a head and 0 otherwise
- If the probability of a head is p then the **likelihood** of a X_i is

$$\mathbb{P}(X_i|p) = p^{X_i}(1 - p)^{1-X_i}$$

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- Suppose we have a coin and we want to establish the probability of a head
- We want to learn this from a series of independent trials
- (Independent trials with two possible outcomes are known in probability theory as Bernoulli trials)
- Let X_i equal 1 if the i^{th} trial is a head and 0 otherwise
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Prior

- We may have a prior belief (e.g. we have made a few trials or we see the coin looks like a normal penny)
- We will suppose we can model our prior belief in terms of a **Beta distribution**

$$f(p) = \text{Beta}(p|a,b) = \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}$$

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Independent Trials

- Using Bayes' rule

$$f(p|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|p) f(p)}{\mathbb{P}(\mathcal{D})}$$

- Assuming the trials are independent (a reasonably fair assumption for tossing coins) then the likelihood factorises

$$\begin{aligned}\mathbb{P}(\mathcal{D}|p) &= \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i} \\ &= p^{X_1} (1-p)^{1-X_1} p^{X_2} (1-p)^{1-X_2} \dots p^{X_n} (1-p)^{1-X_n} \\ &= p^{\sum_i X_i} (1-p)^{\sum_i (1-X_i)} = p^s (1-p)^{n-s}\end{aligned}$$

$$s = \sum_i X_i \text{ (number of successes/heads)}$$

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- The posterior distribution is Beta distribution

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- Something rather nice happened
- Starting with a beta distributed prior $f(p) = \text{Beta}(p|a_0, b_0)$ for a set of Bernoulli trials we obtain a beta distributed posterior $f(p|\mathcal{D}) = \text{Beta}(p|a_0 + s, b_0 + n - s)$
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Incremental Updating

- For independent data we can update incrementally

$$\mathcal{D} = (X_1, X_2, \dots, X_n)$$

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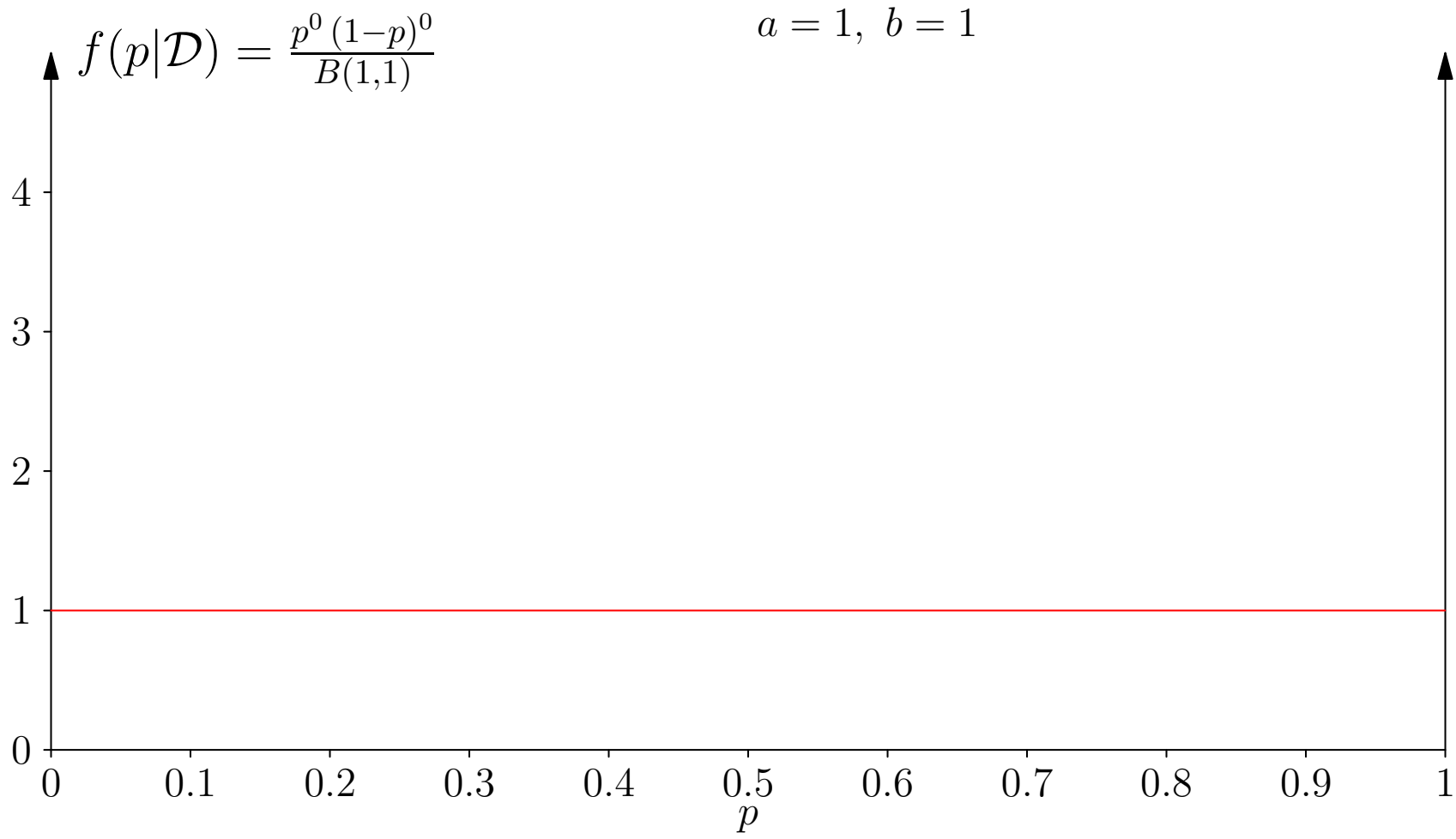
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Example ($p=0.7$)

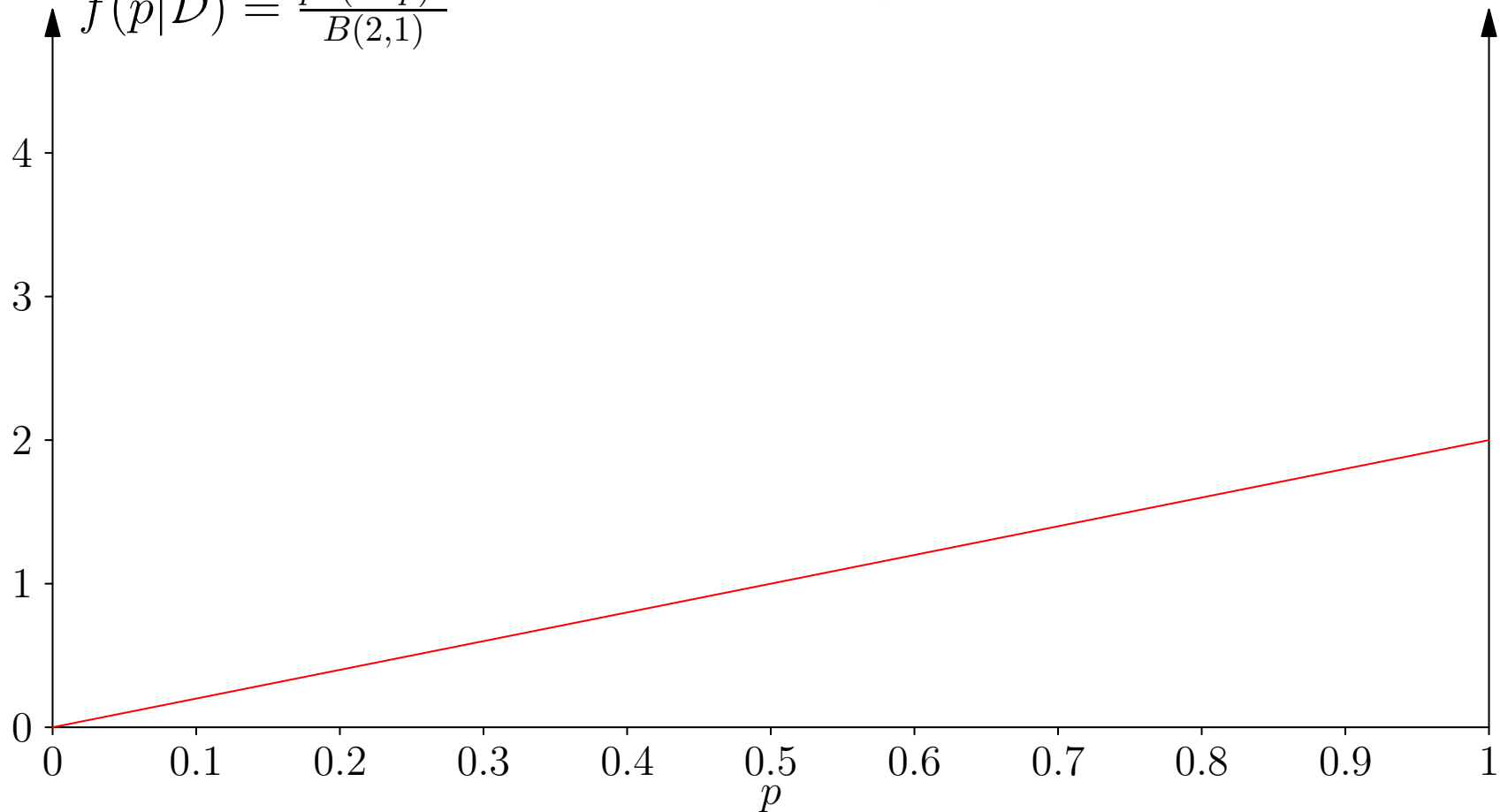


Example (p=0.7)

$$\mathcal{D} = \{H,$$

$$f(p|\mathcal{D}) = \frac{p^1 (1-p)^0}{B(2,1)}$$

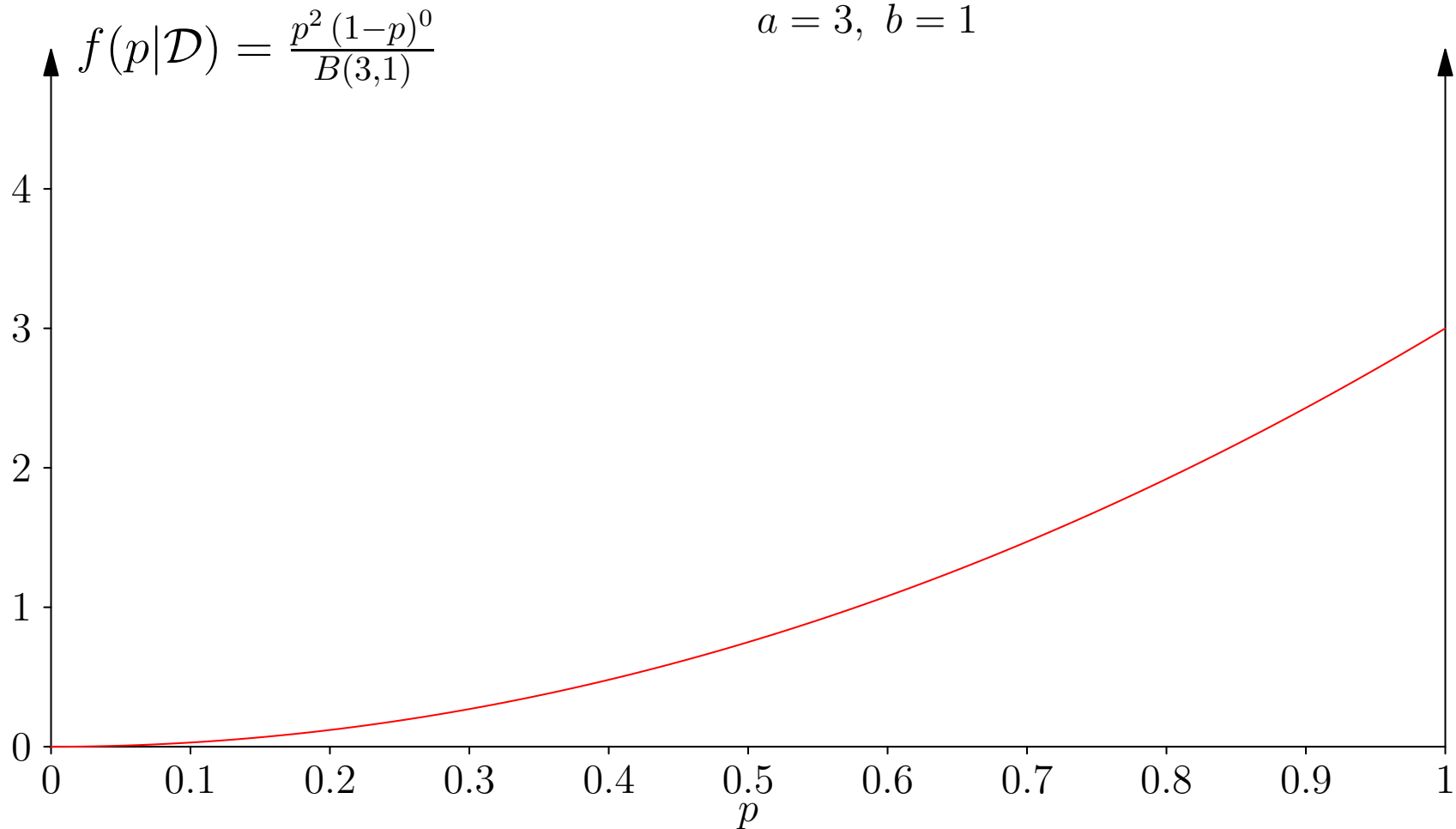
$$a = 2, \quad b = 1$$



Example (p=0.7)

$$\mathcal{D} = \{H, H,$$

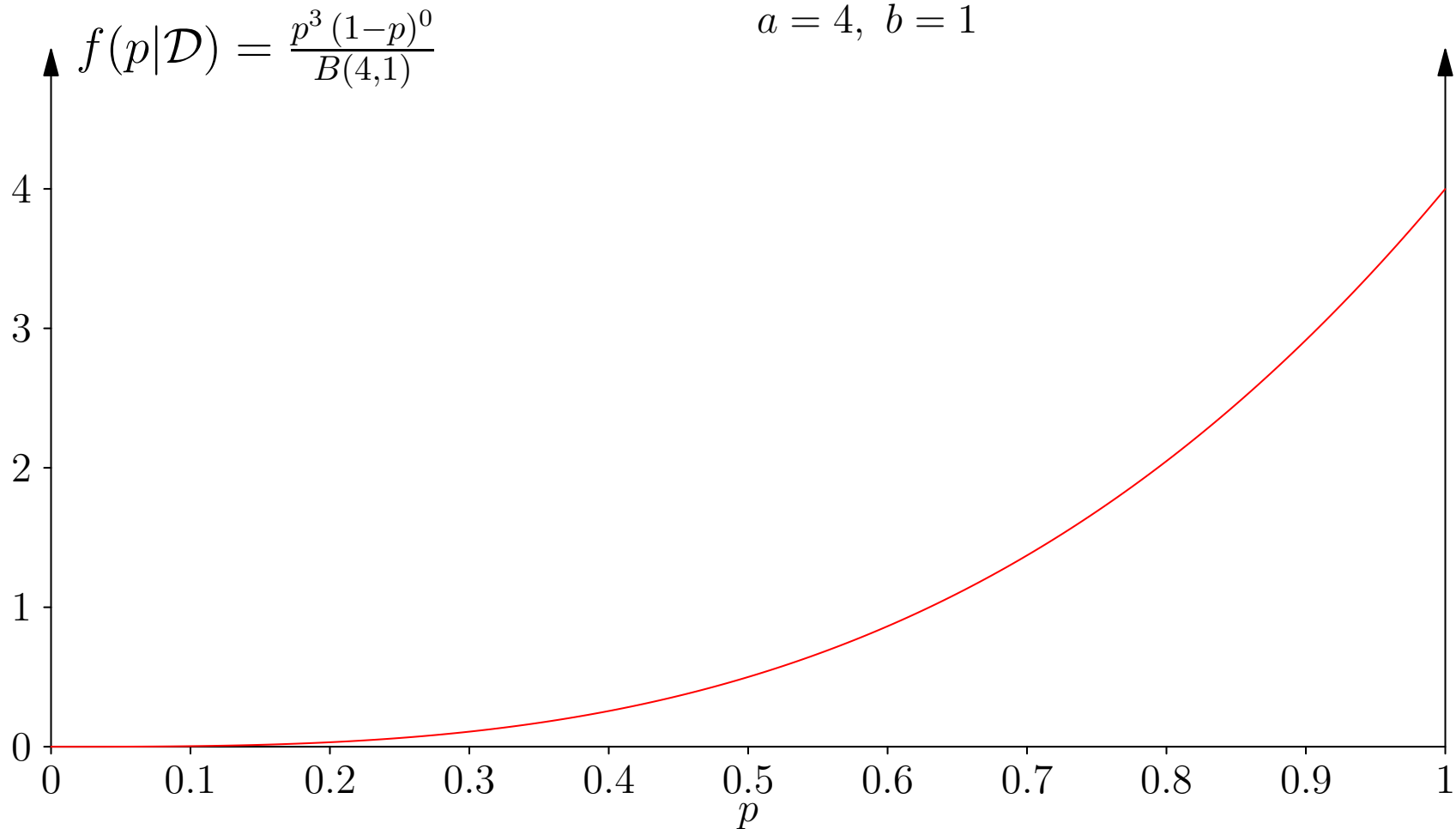
$$a = 3, b = 1$$



Example (p=0.7)

$$\mathcal{D} = \{H, H, H,$$

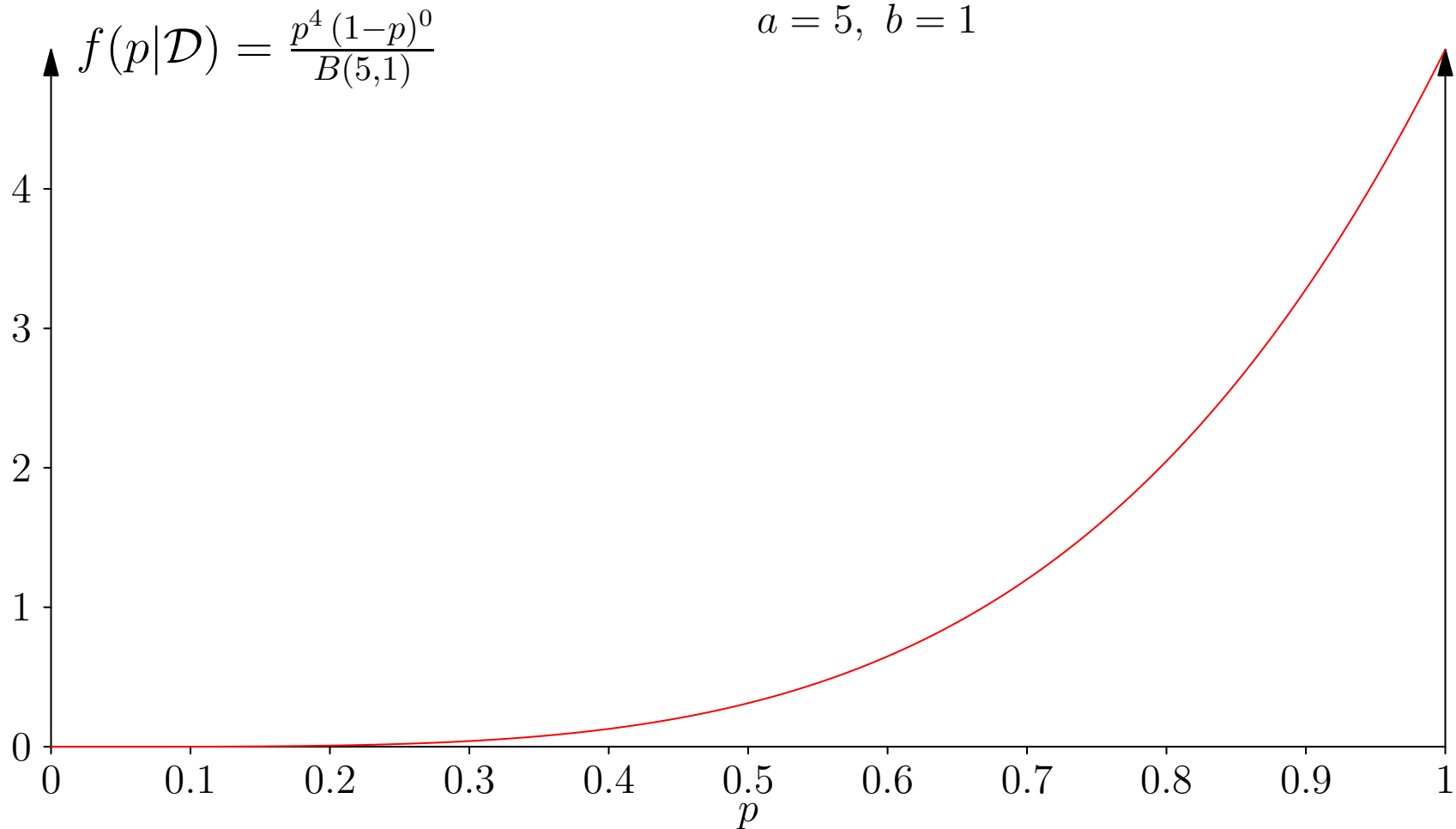
$$a = 4, b = 1$$



Example (p=0.7)

$$\mathcal{D} = \{H, H, H, H,$$

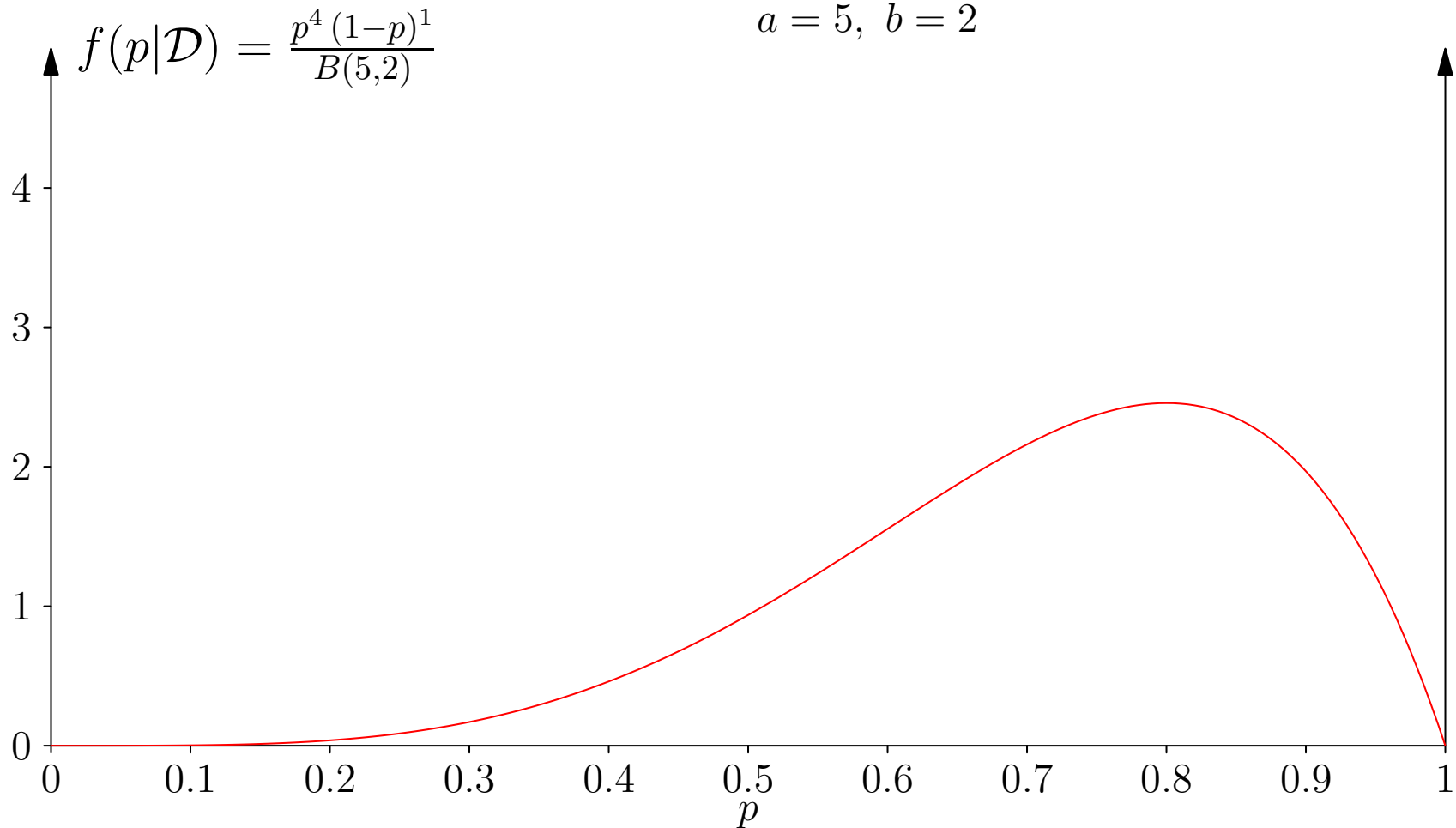
$$a = 5, b = 1$$



Example (p=0.7)

$$\mathcal{D} = \{H, H, H, H, T,$$

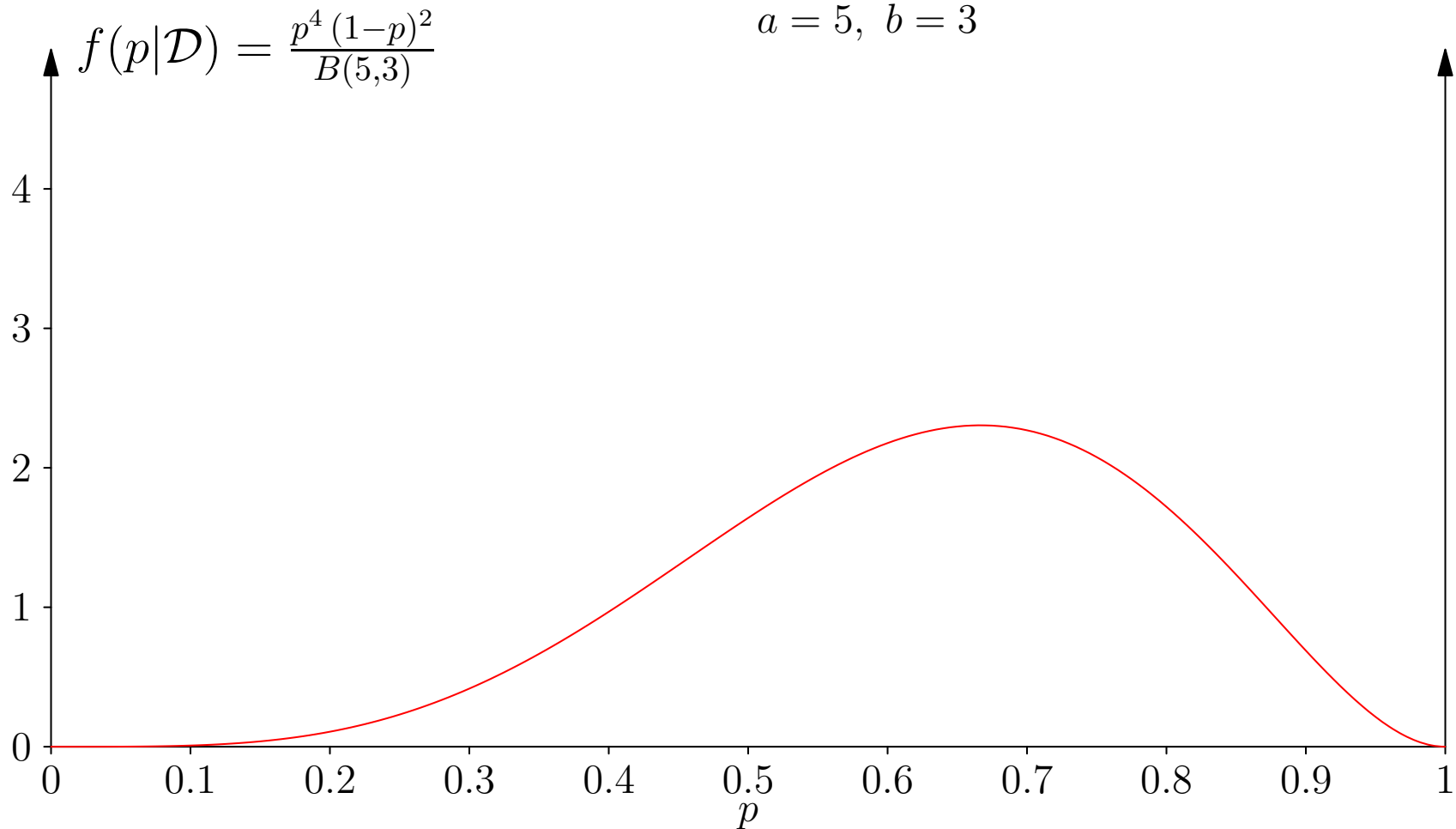
$$a = 5, \quad b = 2$$



Example ($p=0.7$)

$$\mathcal{D} = \{H, H, H, H, T, T,$$

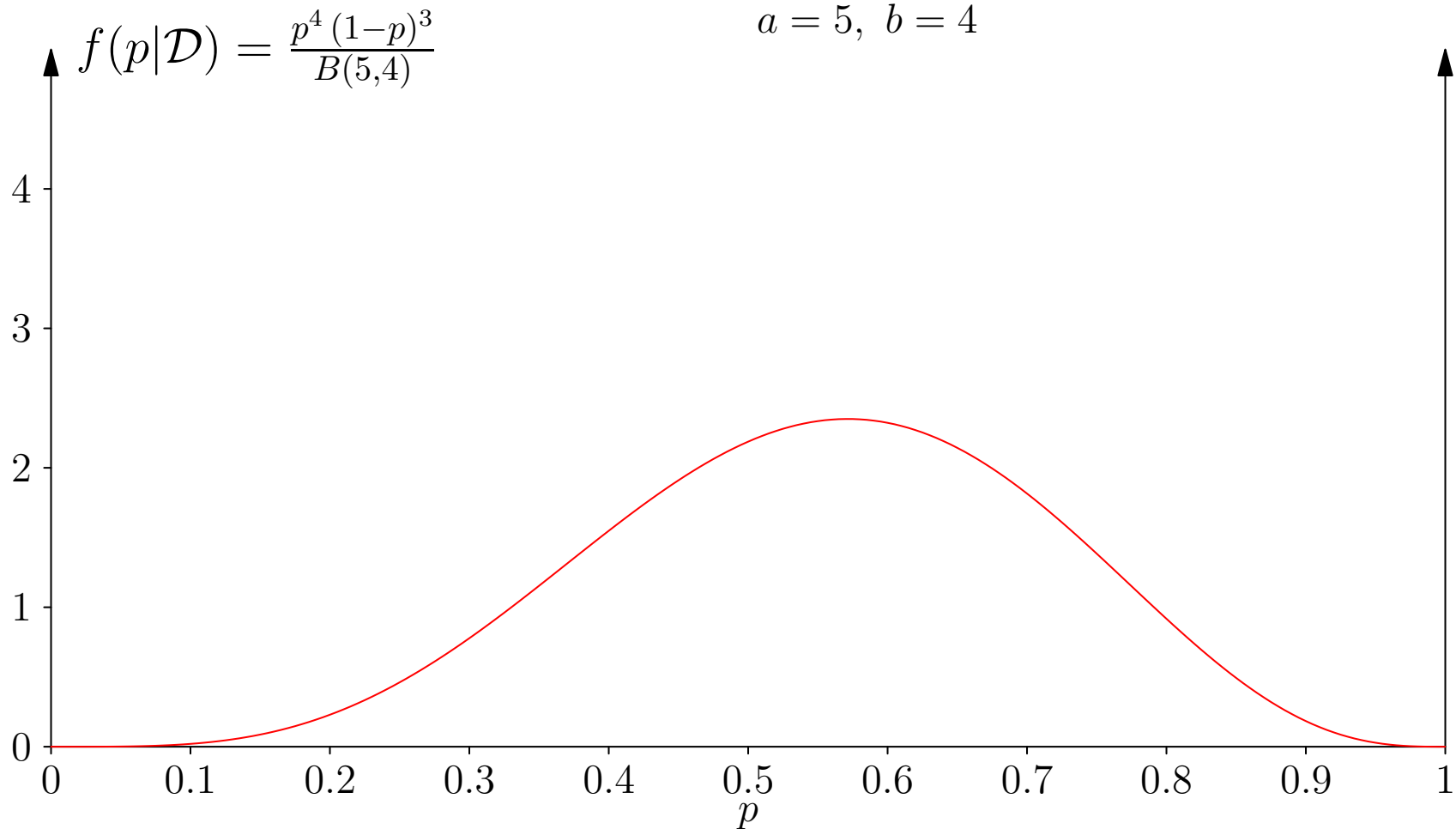
$$a = 5, \quad b = 3$$



Example ($p=0.7$)

$$\mathcal{D} = \{H, H, H, H, T, T, T,$$

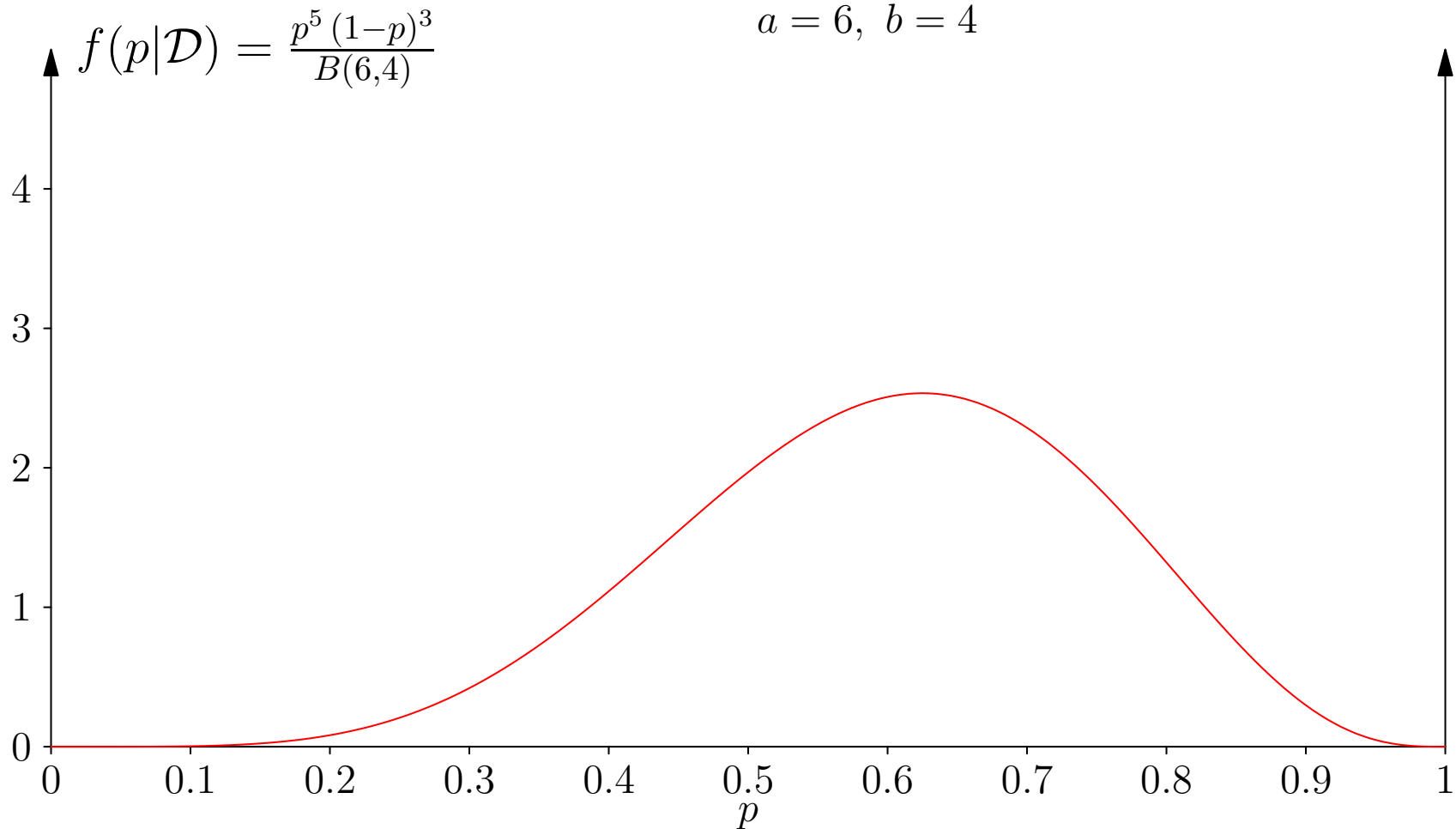
$$a = 5, \quad b = 4$$



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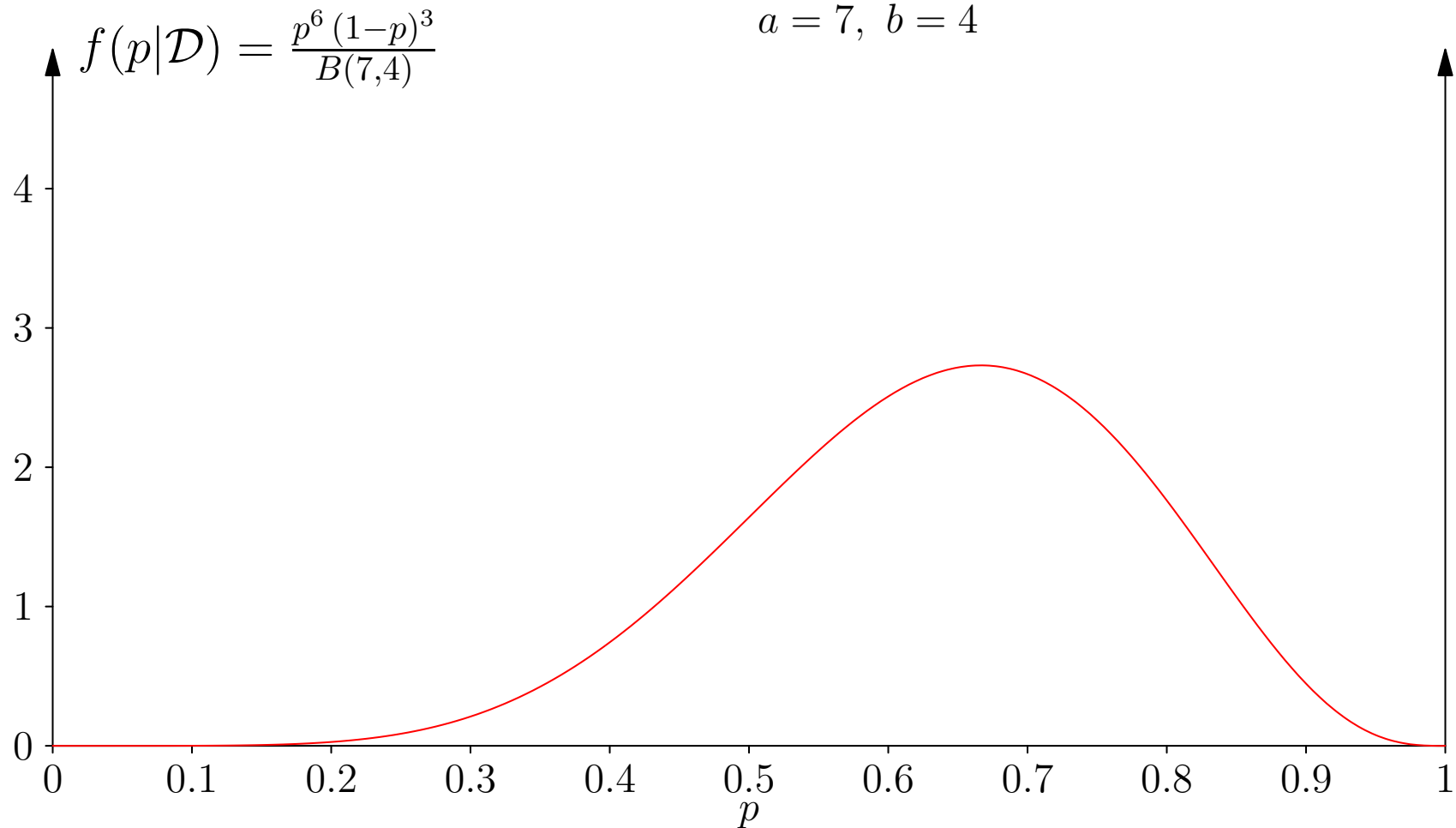
$$a = 6, \quad b = 4$$



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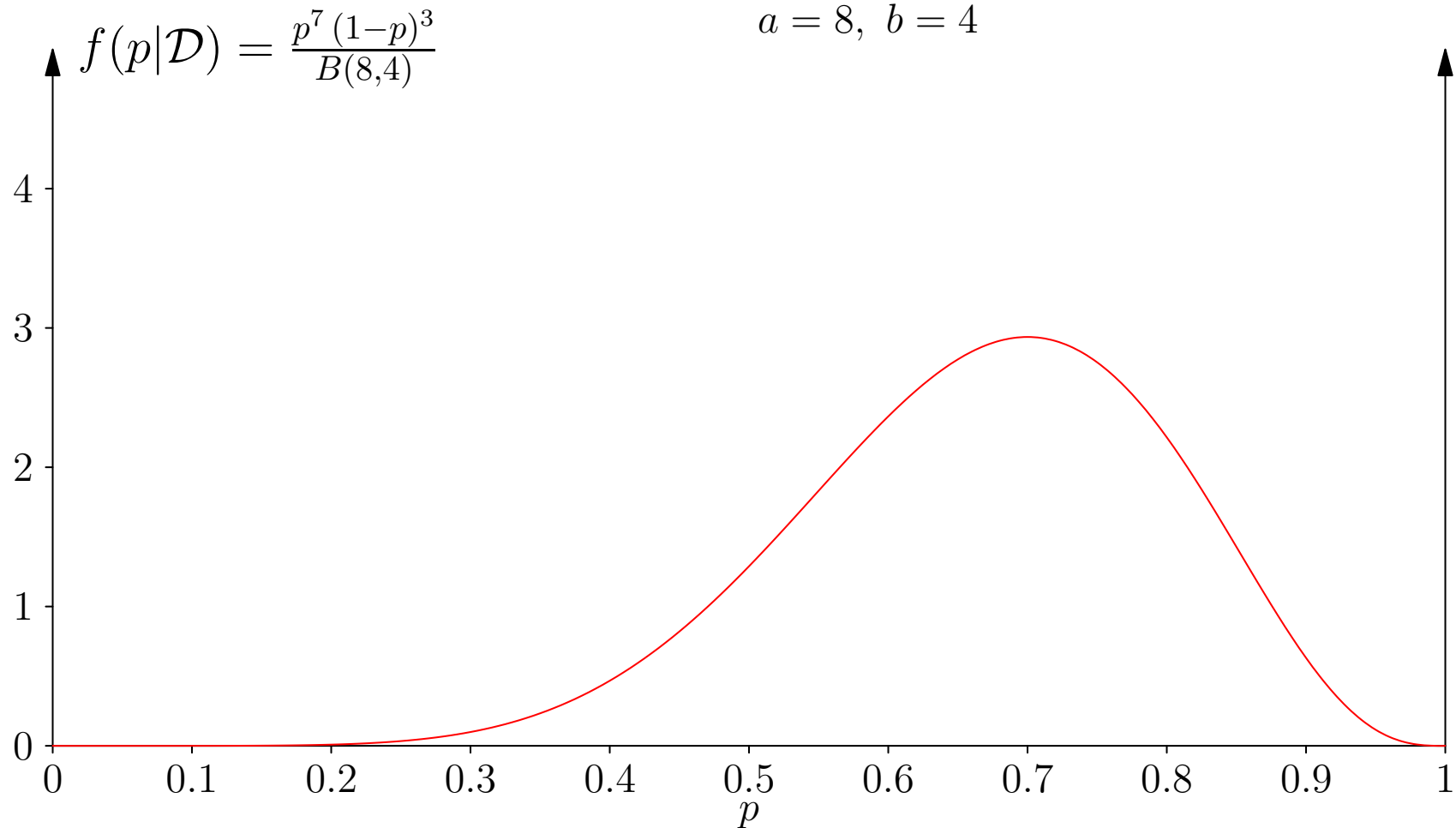
$$a = 7, \quad b = 4$$



Example ($p=0.7$)

$$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H,$$

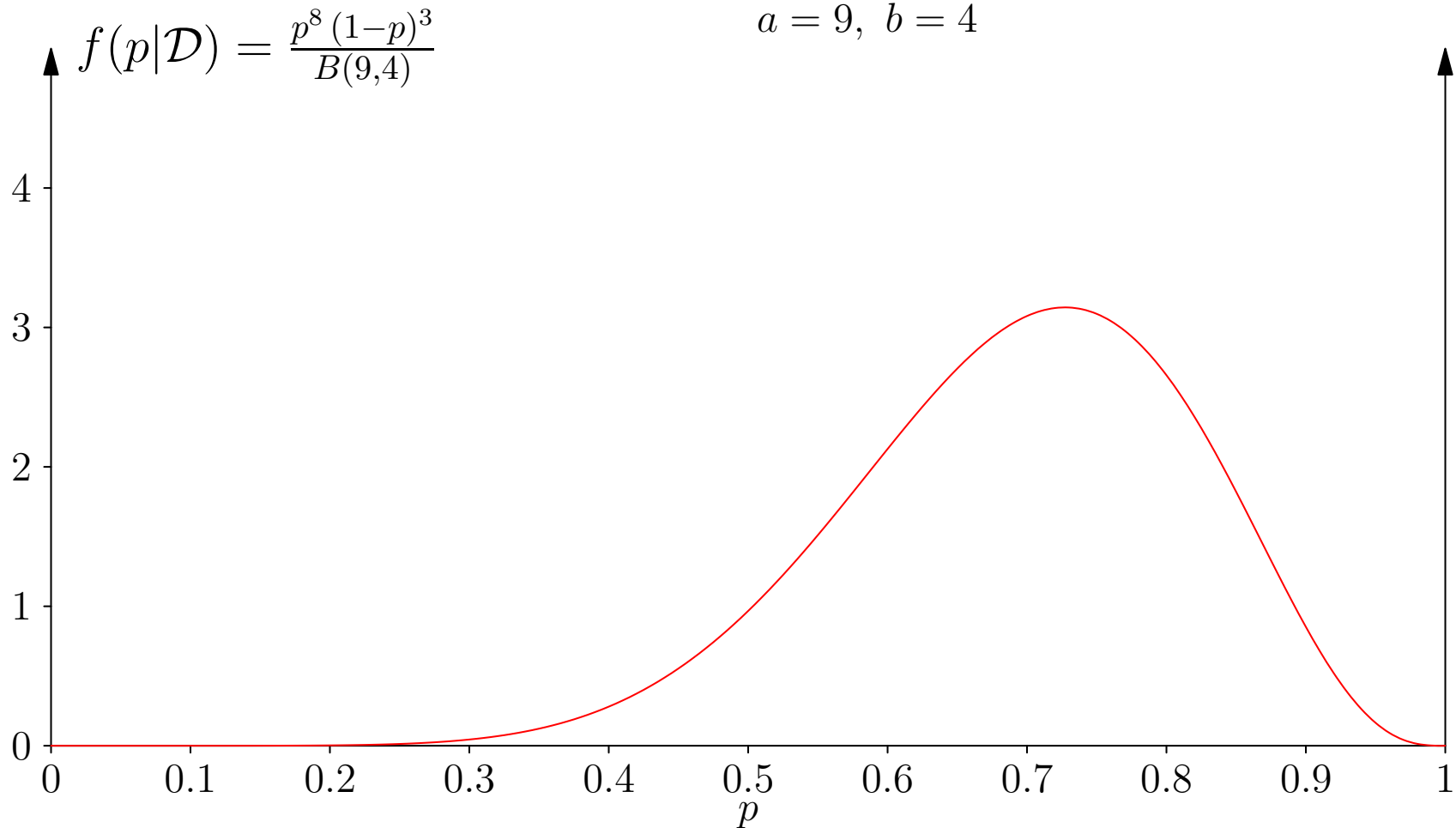
$$a = 8, \quad b = 4$$



Example ($p=0.7$)

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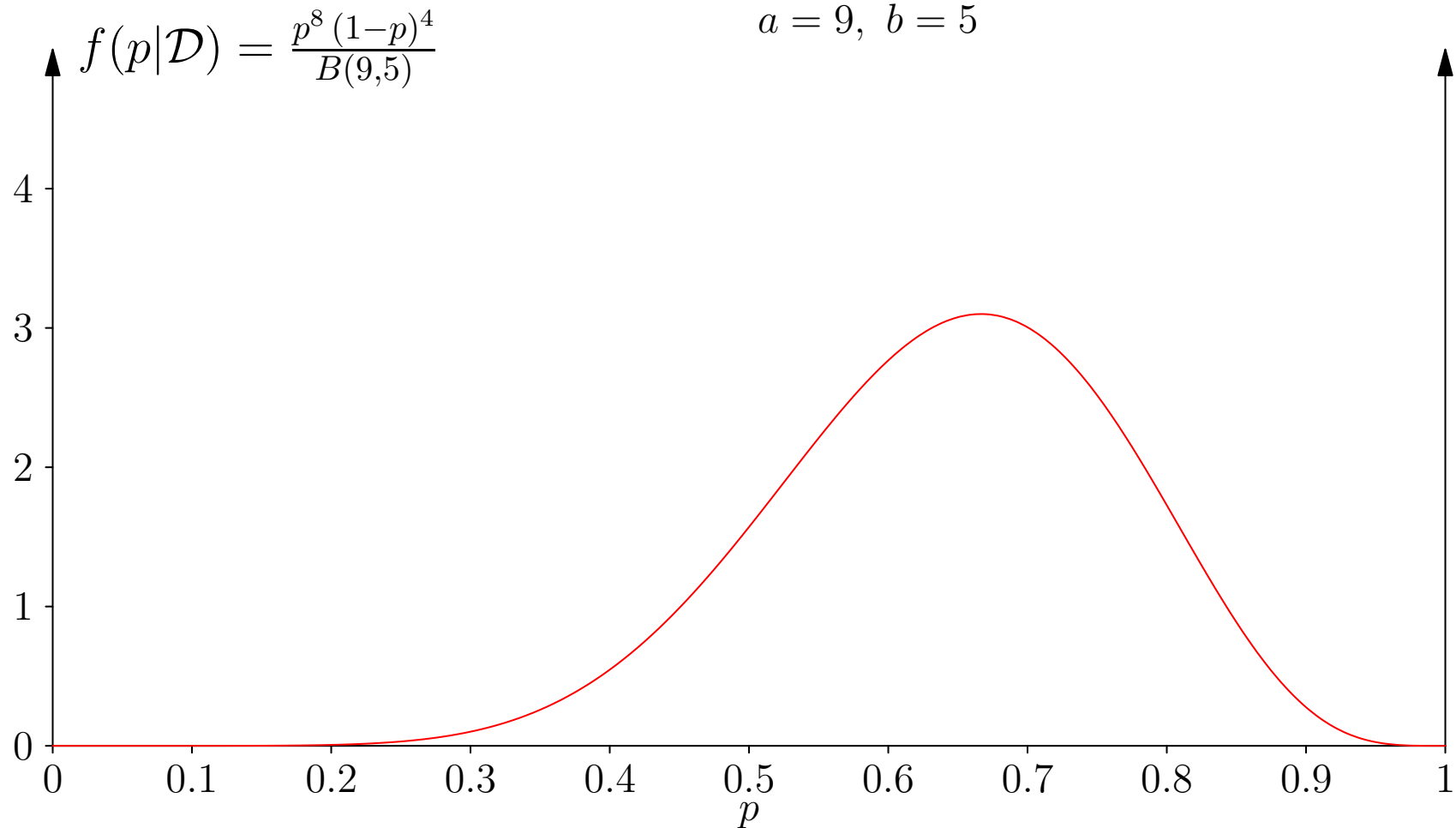
$a = 9, b = 4$



Example ($p=0.7$)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T,$

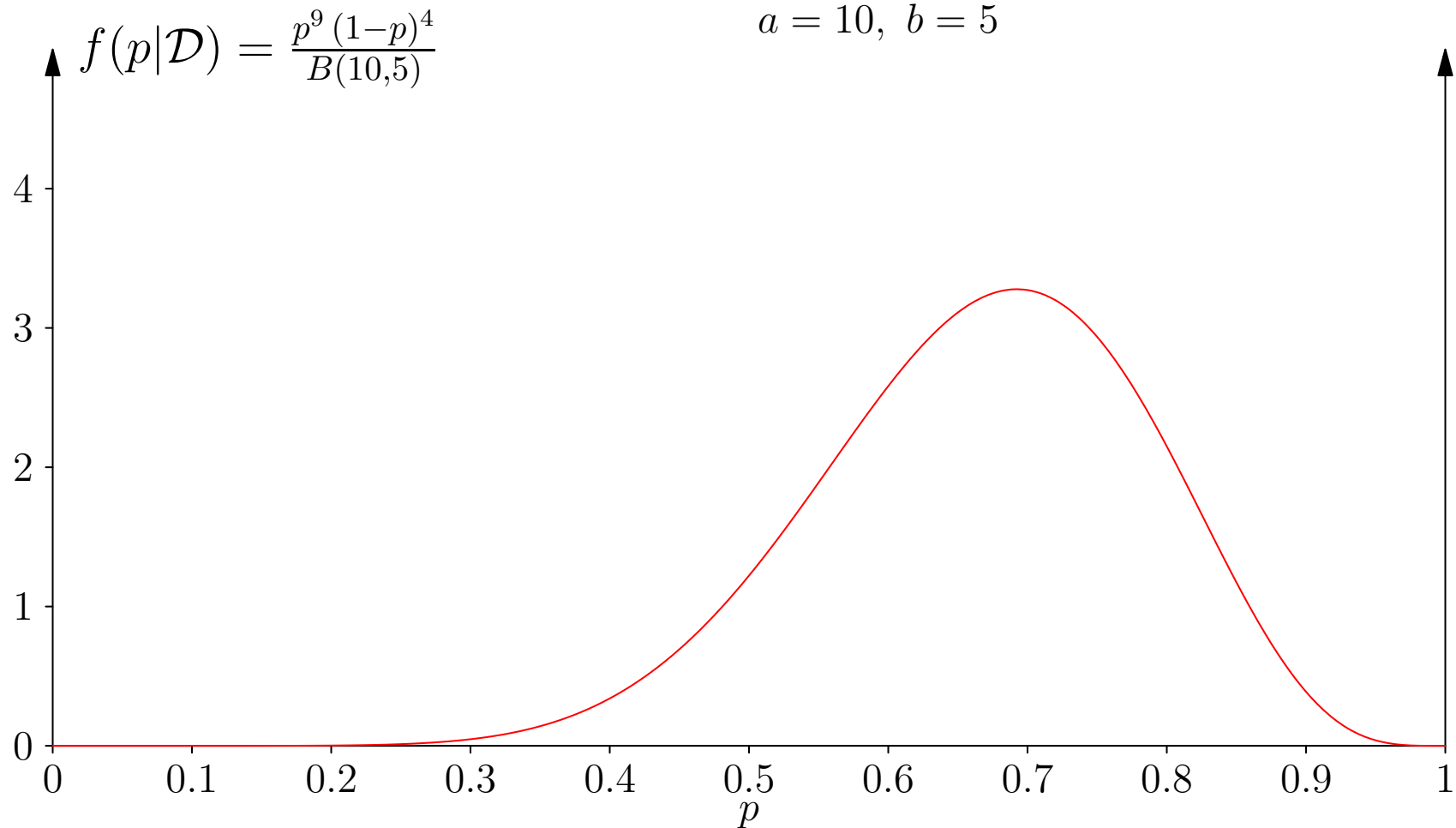
$a = 9, b = 5$



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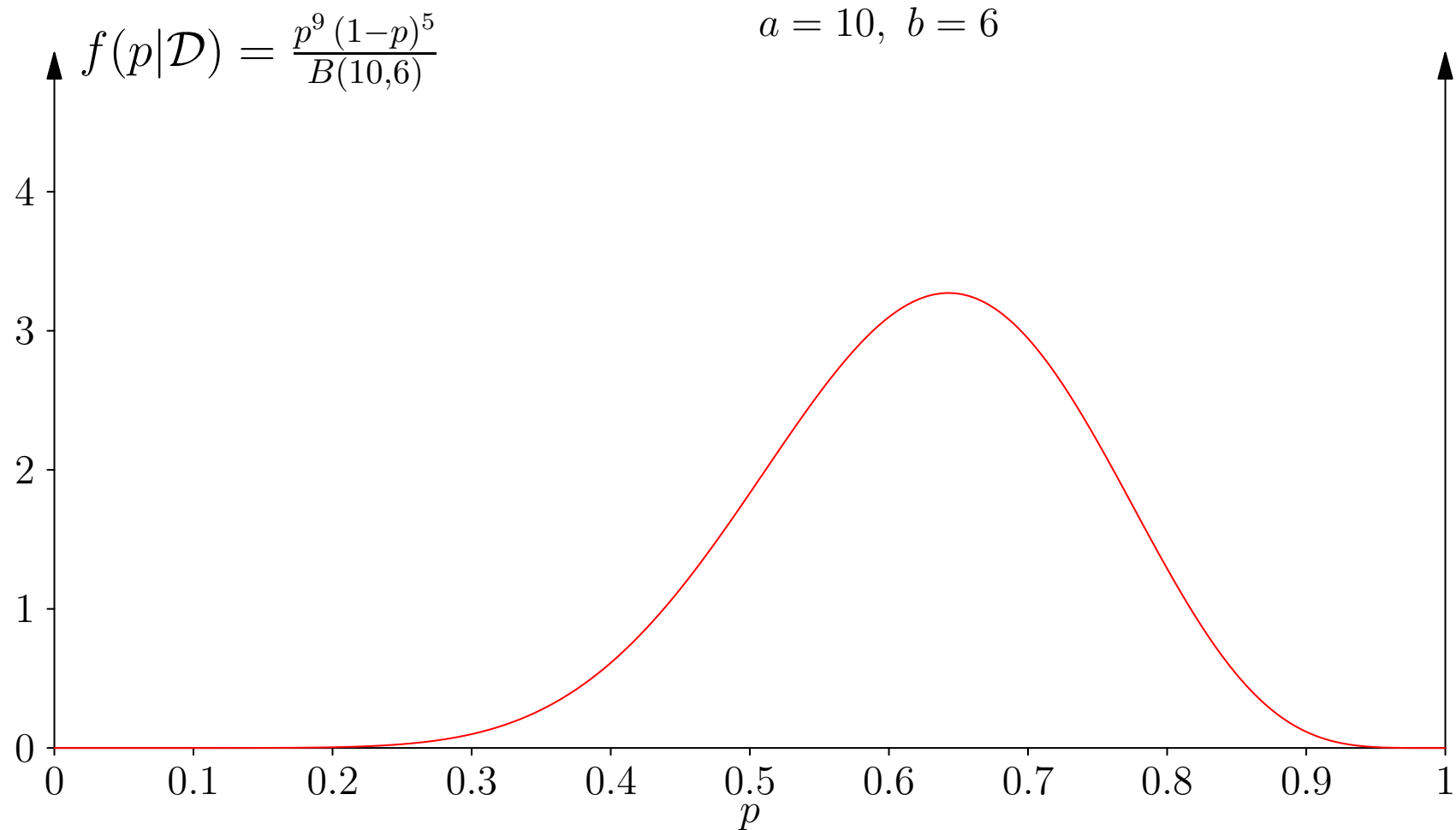
$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H,$

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Example ($p=0.7$)

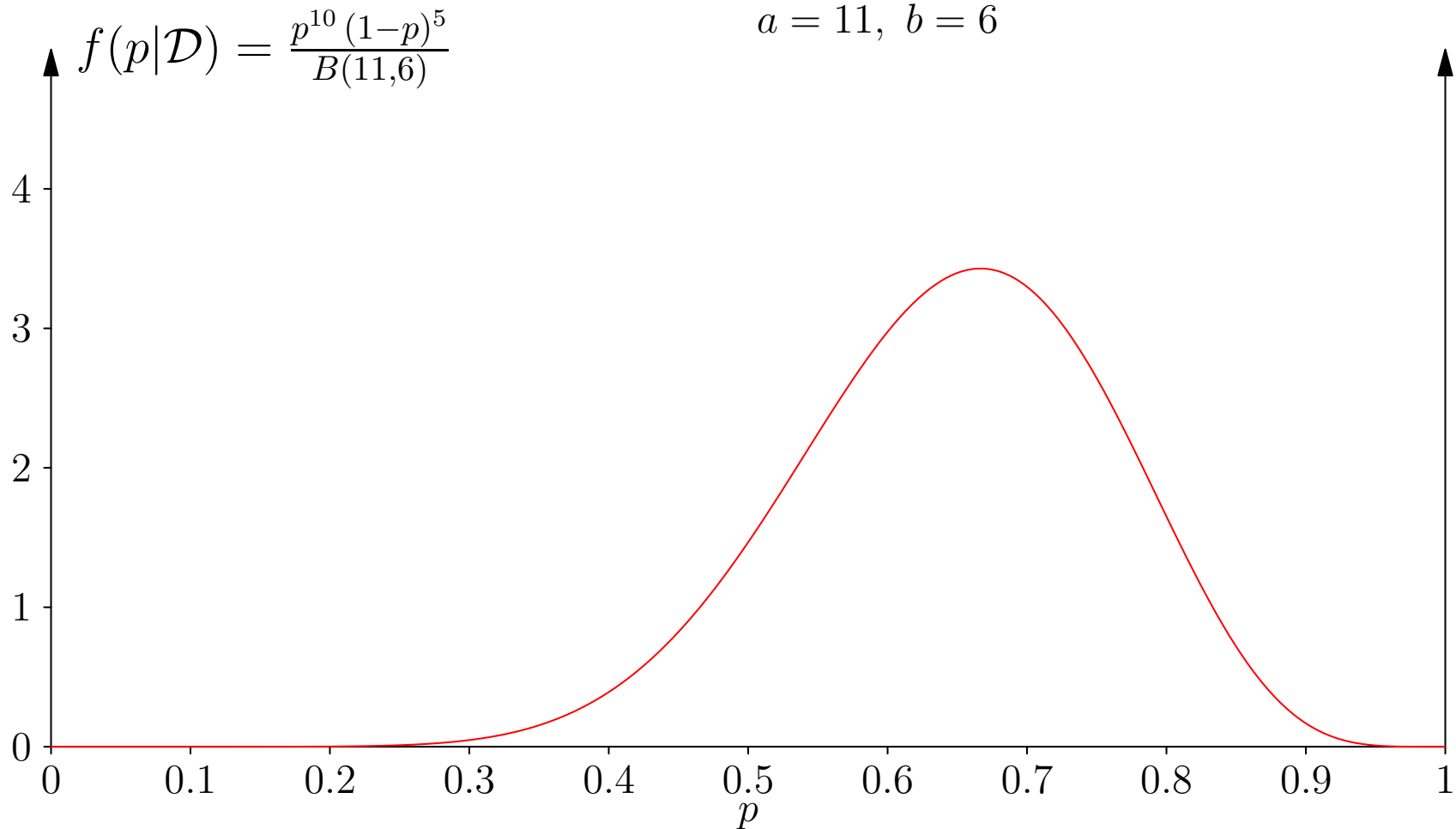
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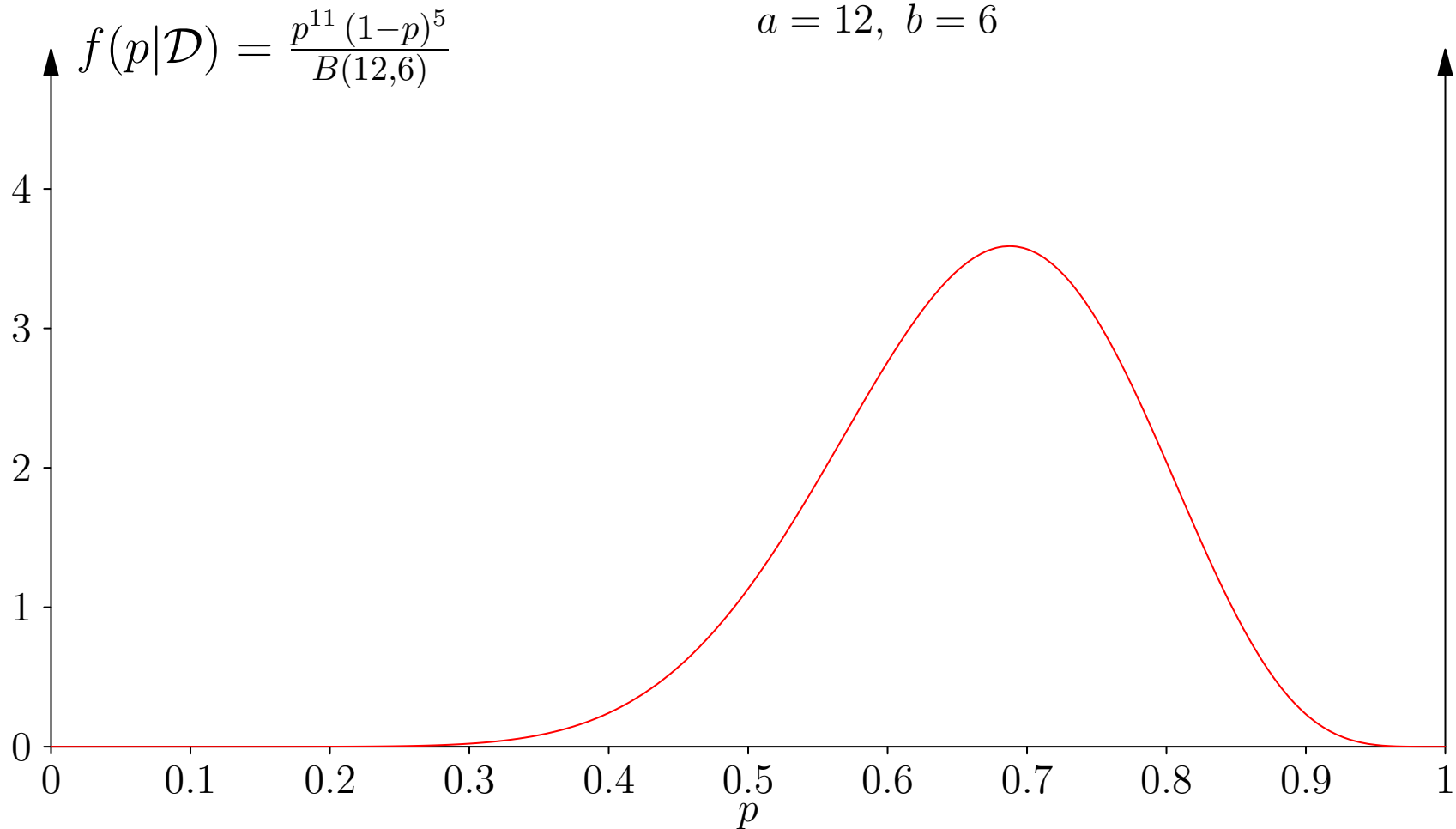
$a = 11, b = 6$



Example ($p=0.7$)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H,$

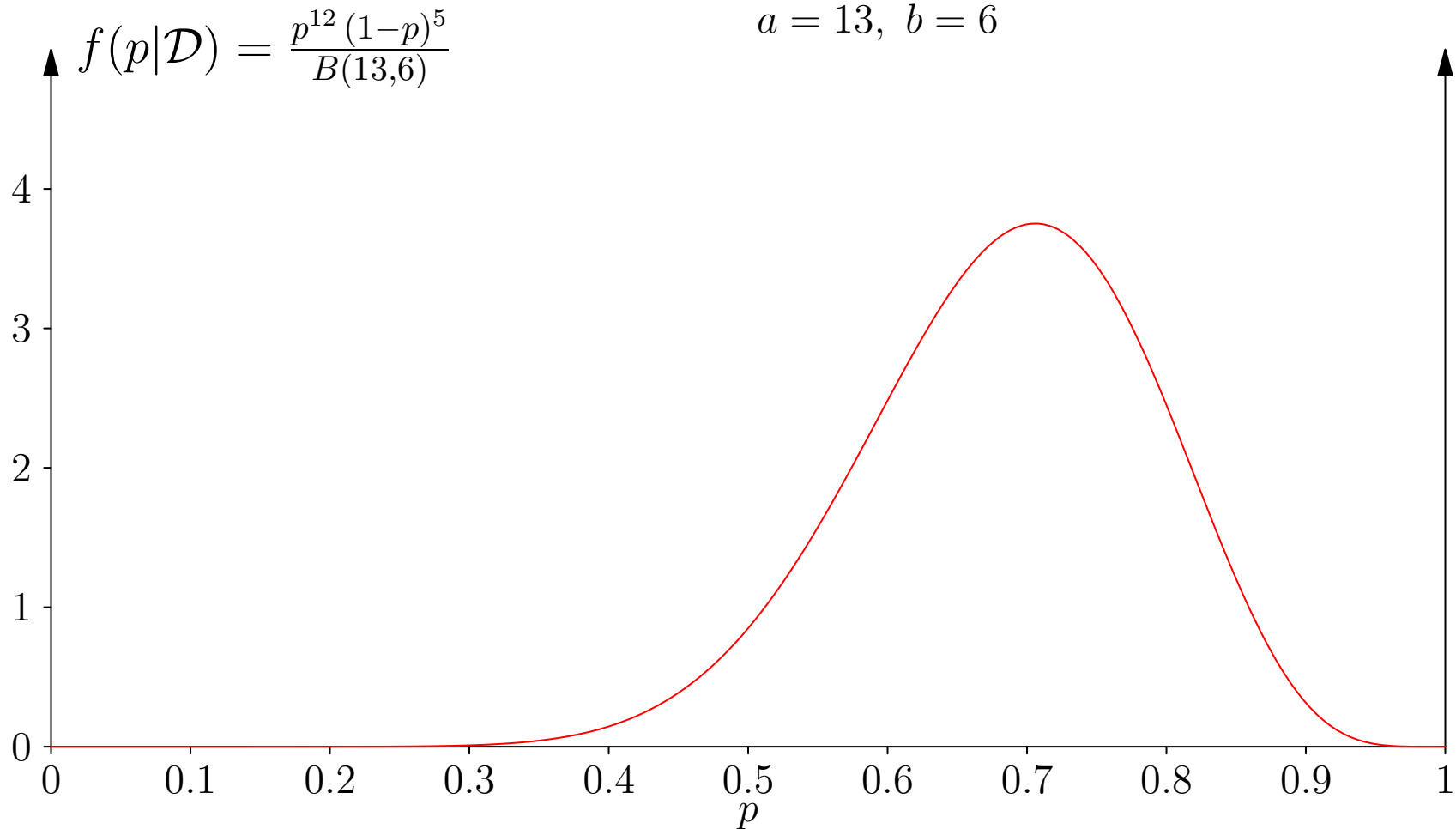
$a = 12, b = 6$



Example (p=0.7)

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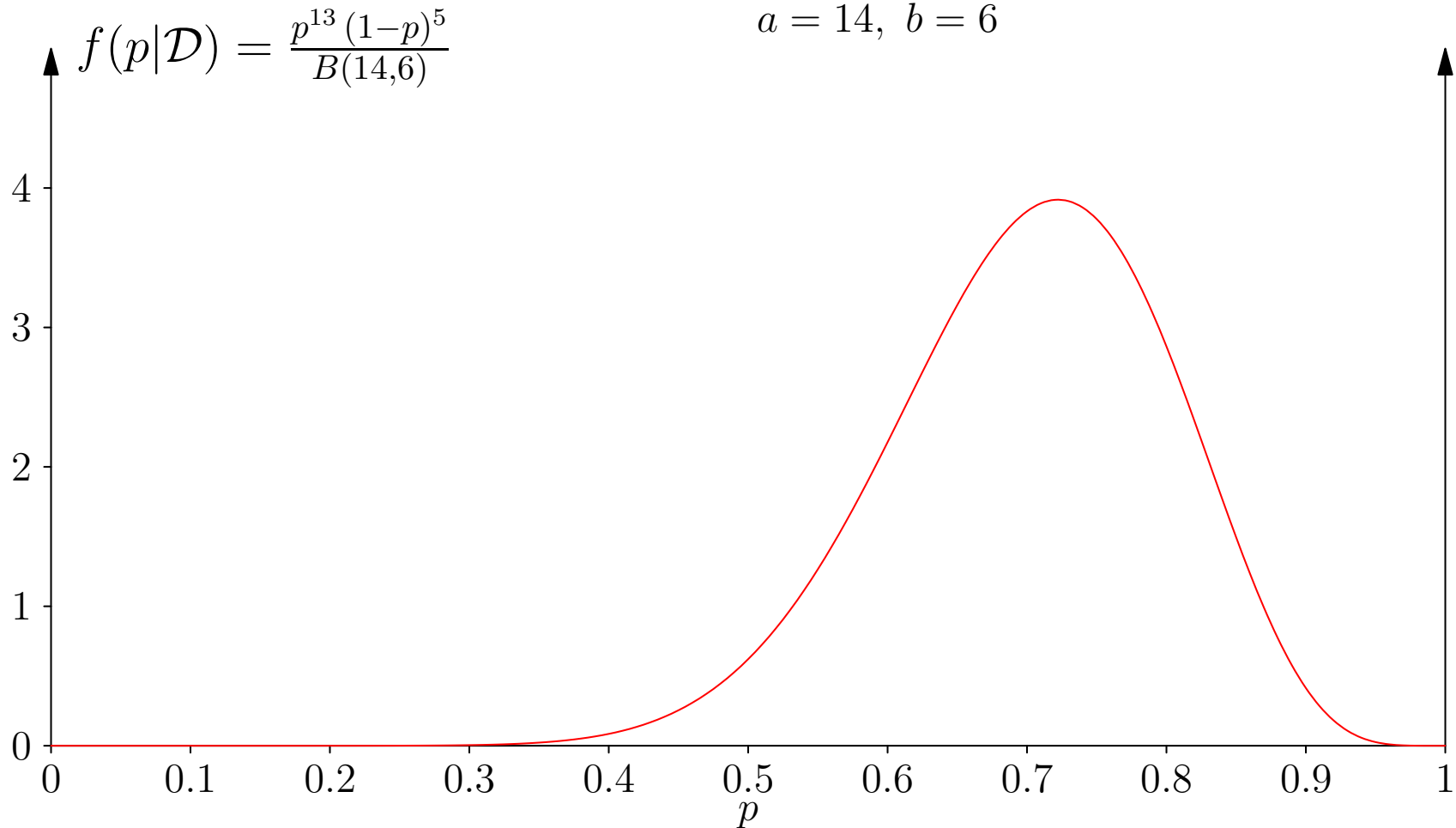
$a = 13, b = 6$



Example (p=0.7)

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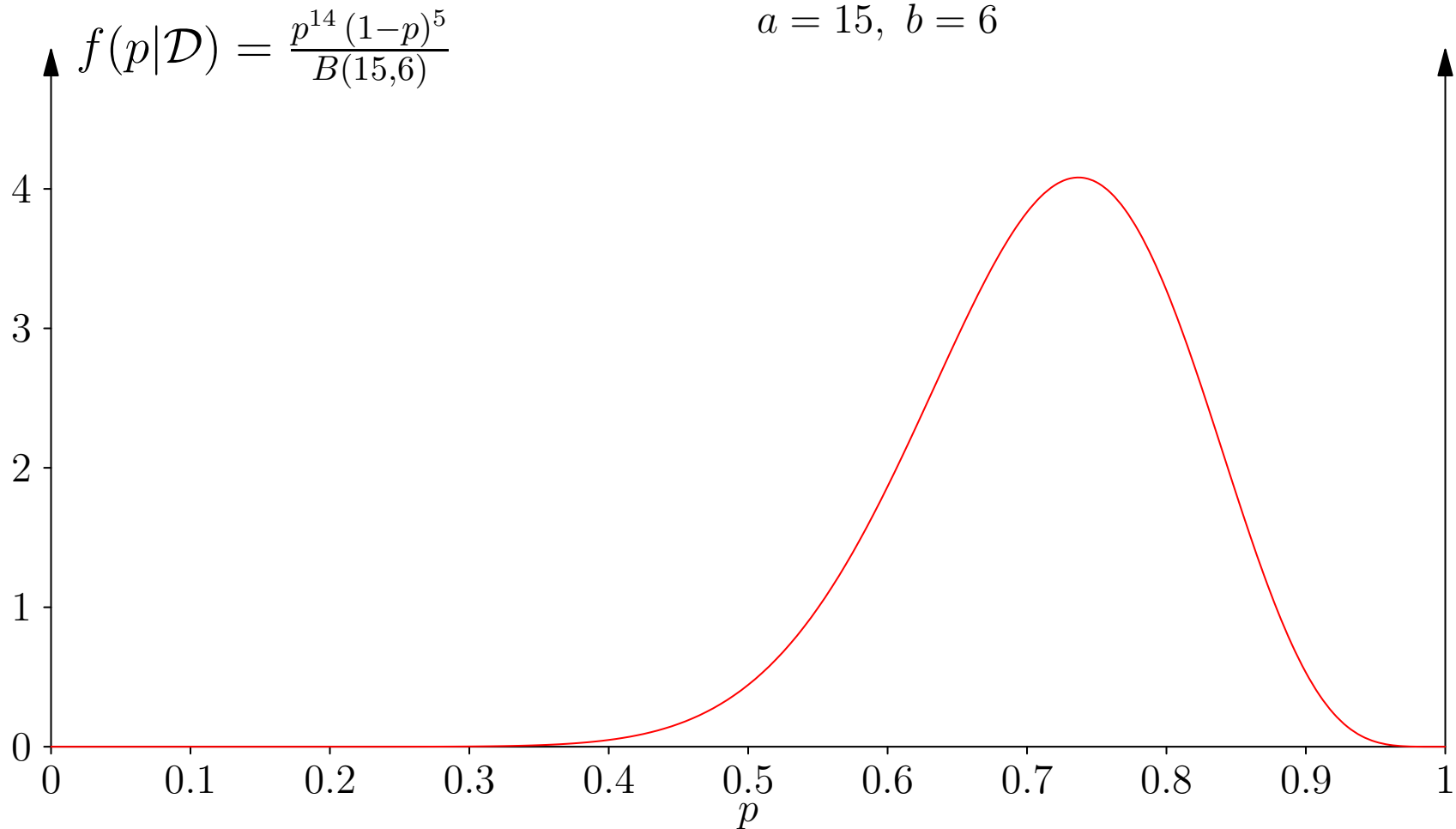
$a = 14, b = 6$



Example ($p=0.7$)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H,$

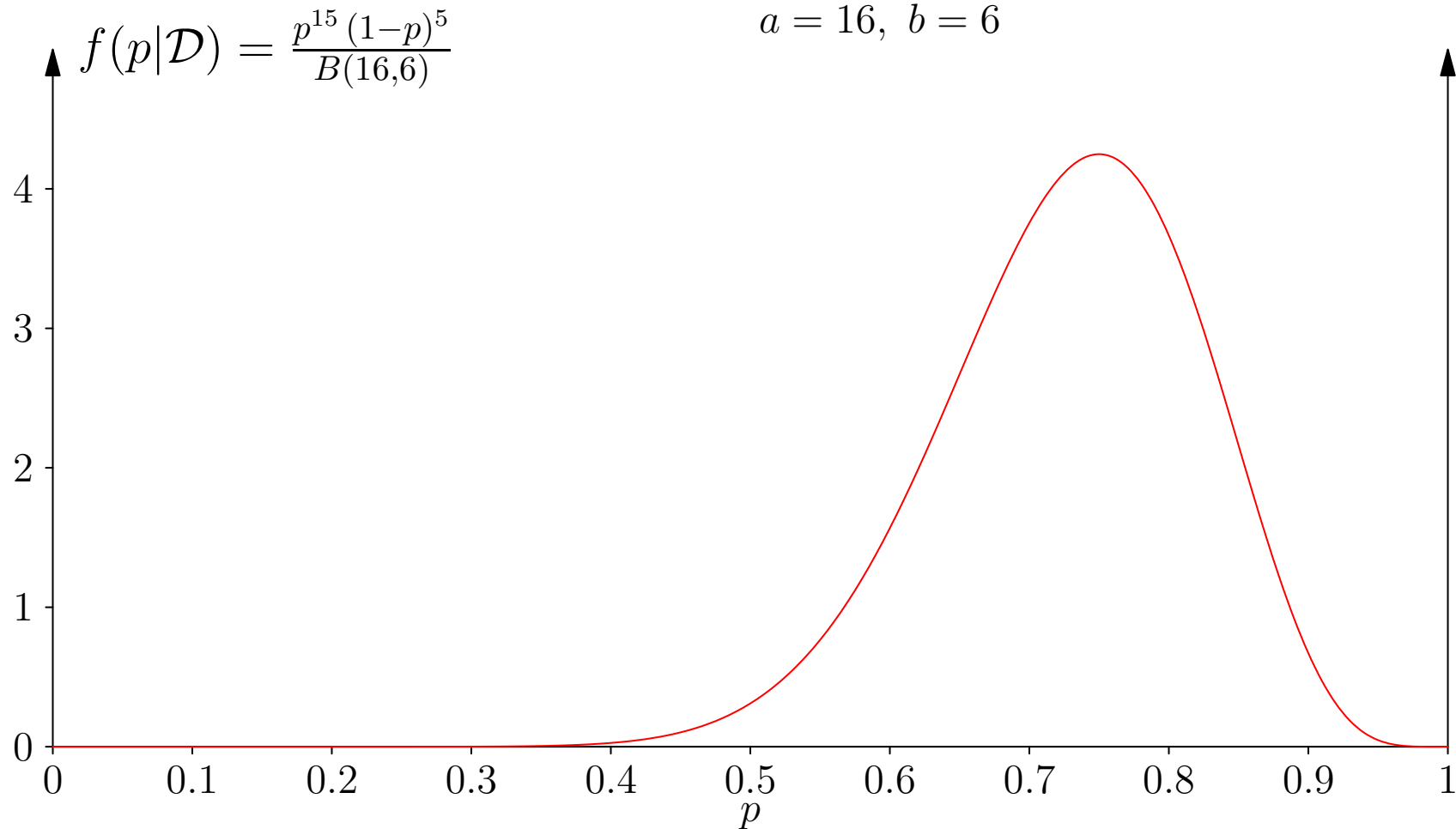
$a = 15, b = 6$



Example ($p=0.7$)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H,$

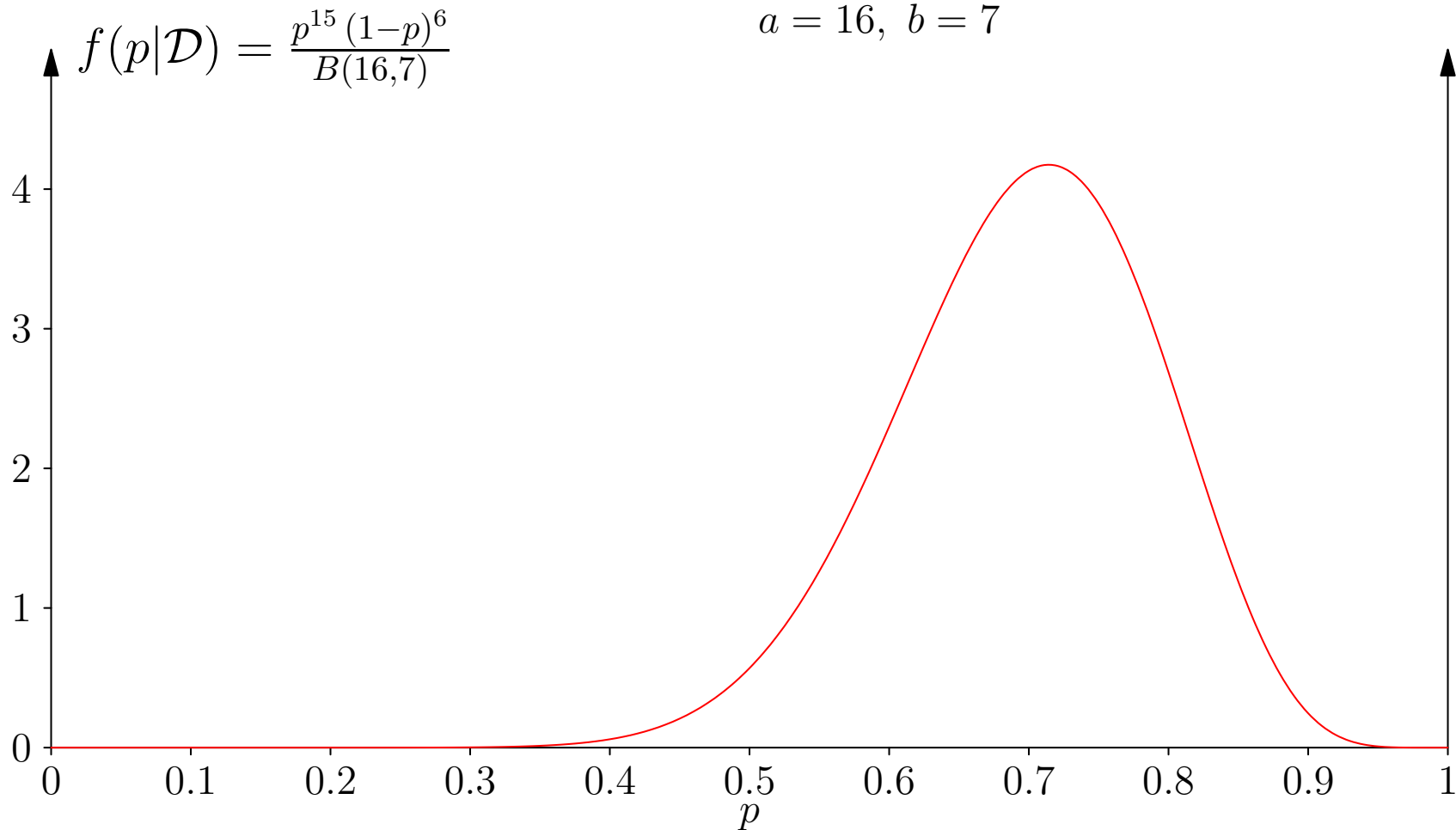
$a = 16, b = 6$



Example (p=0.7)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H, T,$

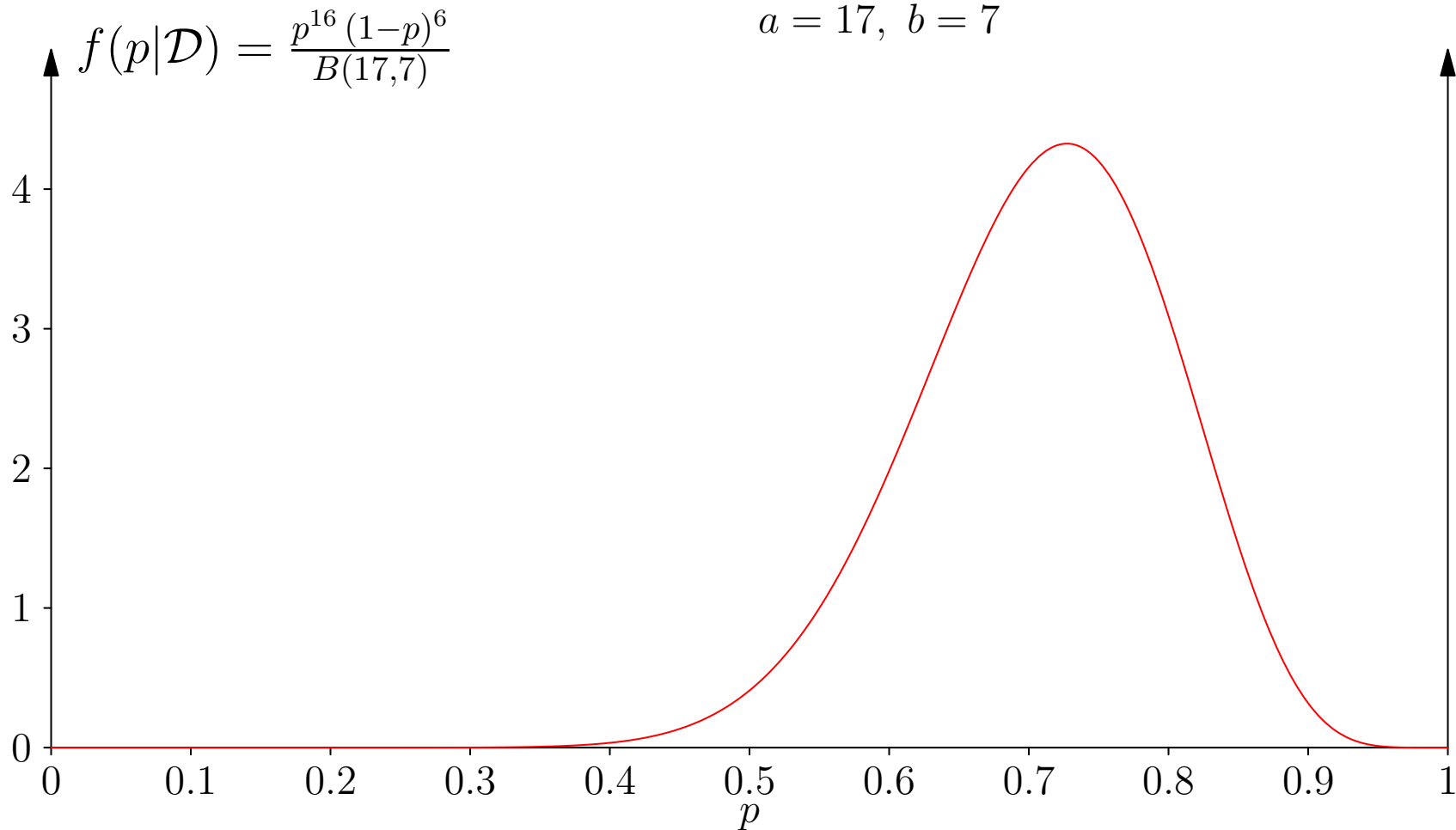
$a = 16, b = 7$



Example (p=0.7)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H, T, H\}$

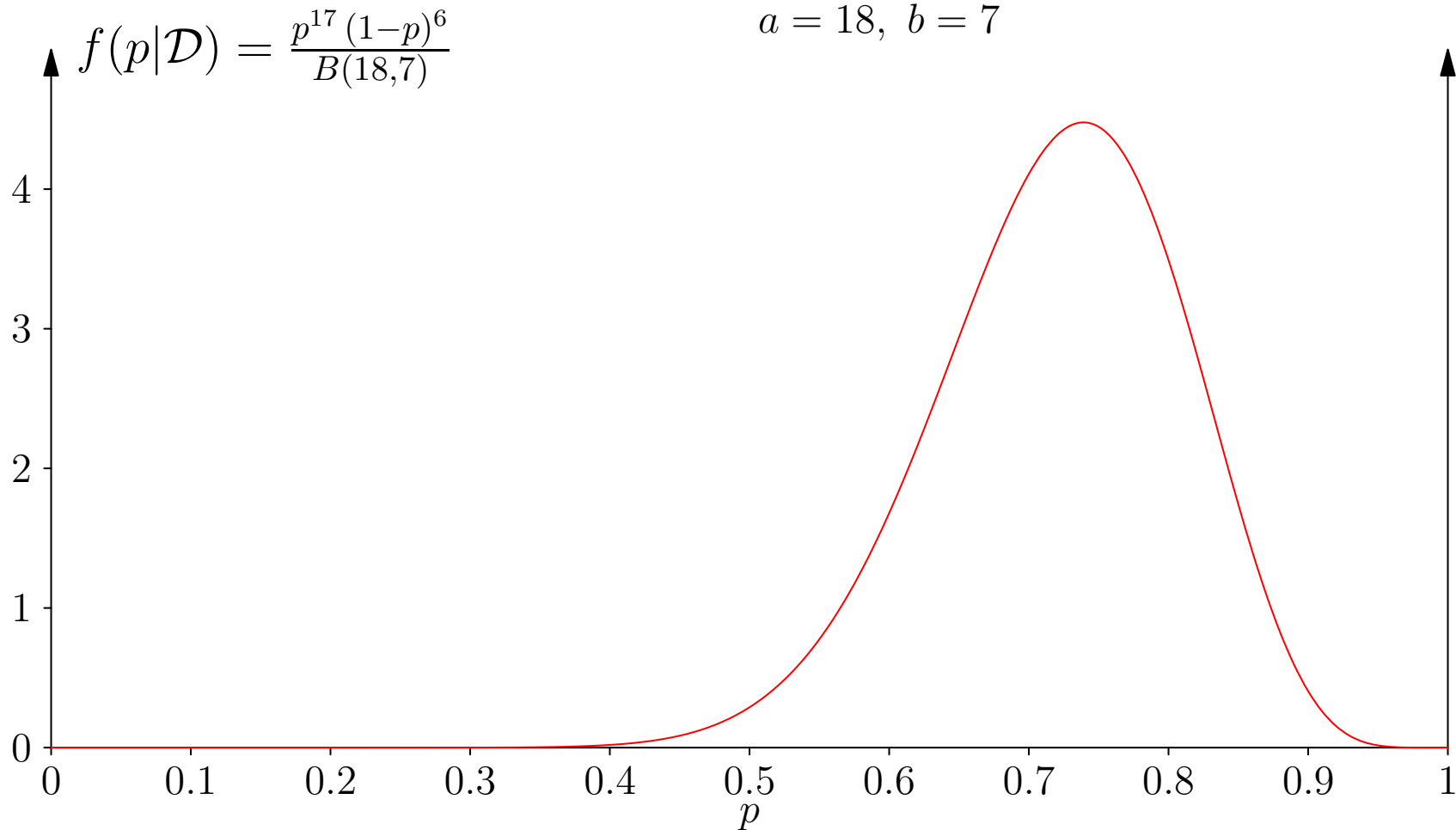
$a = 17, b = 7$



Example ($p=0.7$)

$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H, T, H, H,$

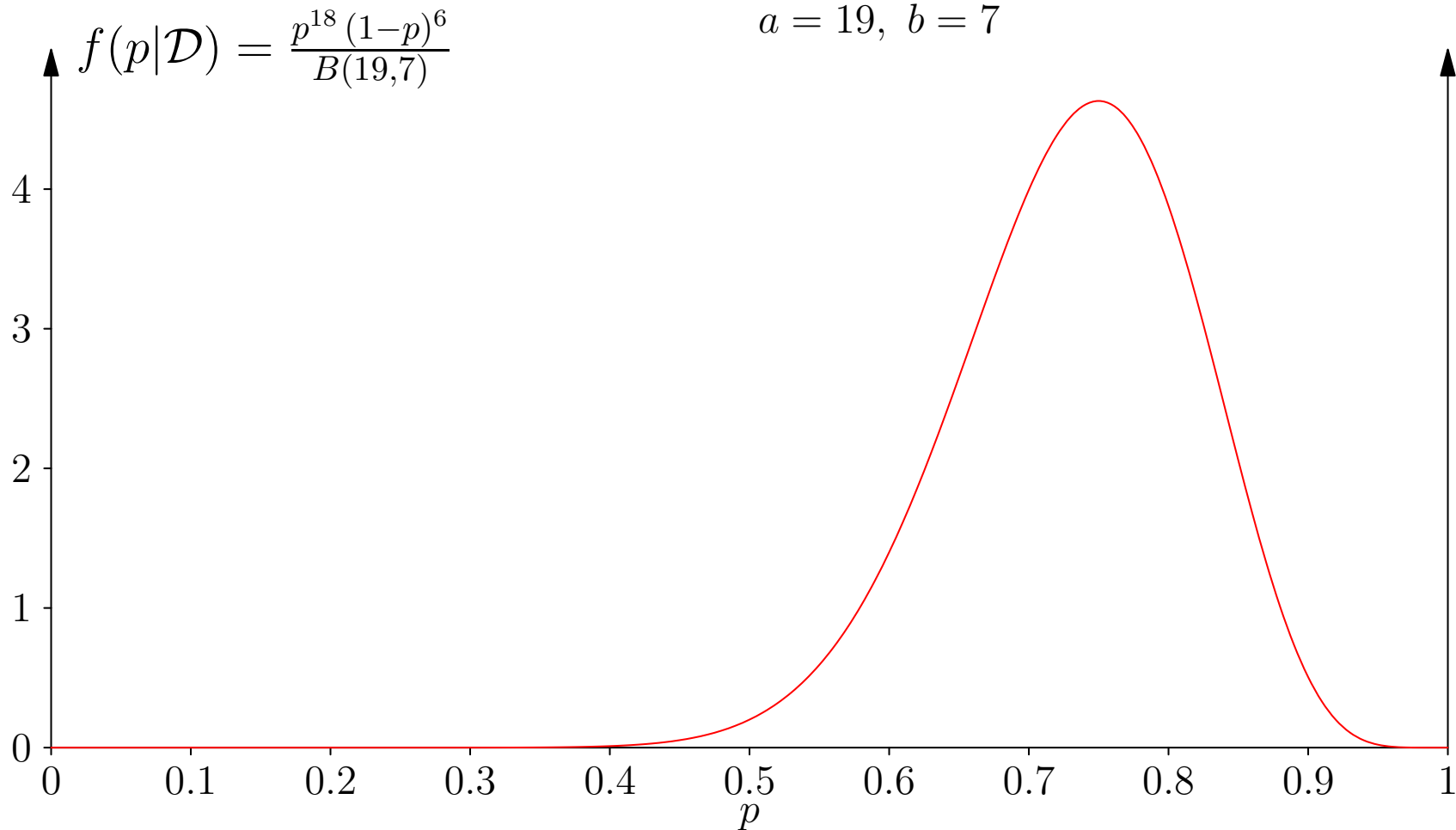
$a = 18, b = 7$



Example ($p=0.7$)

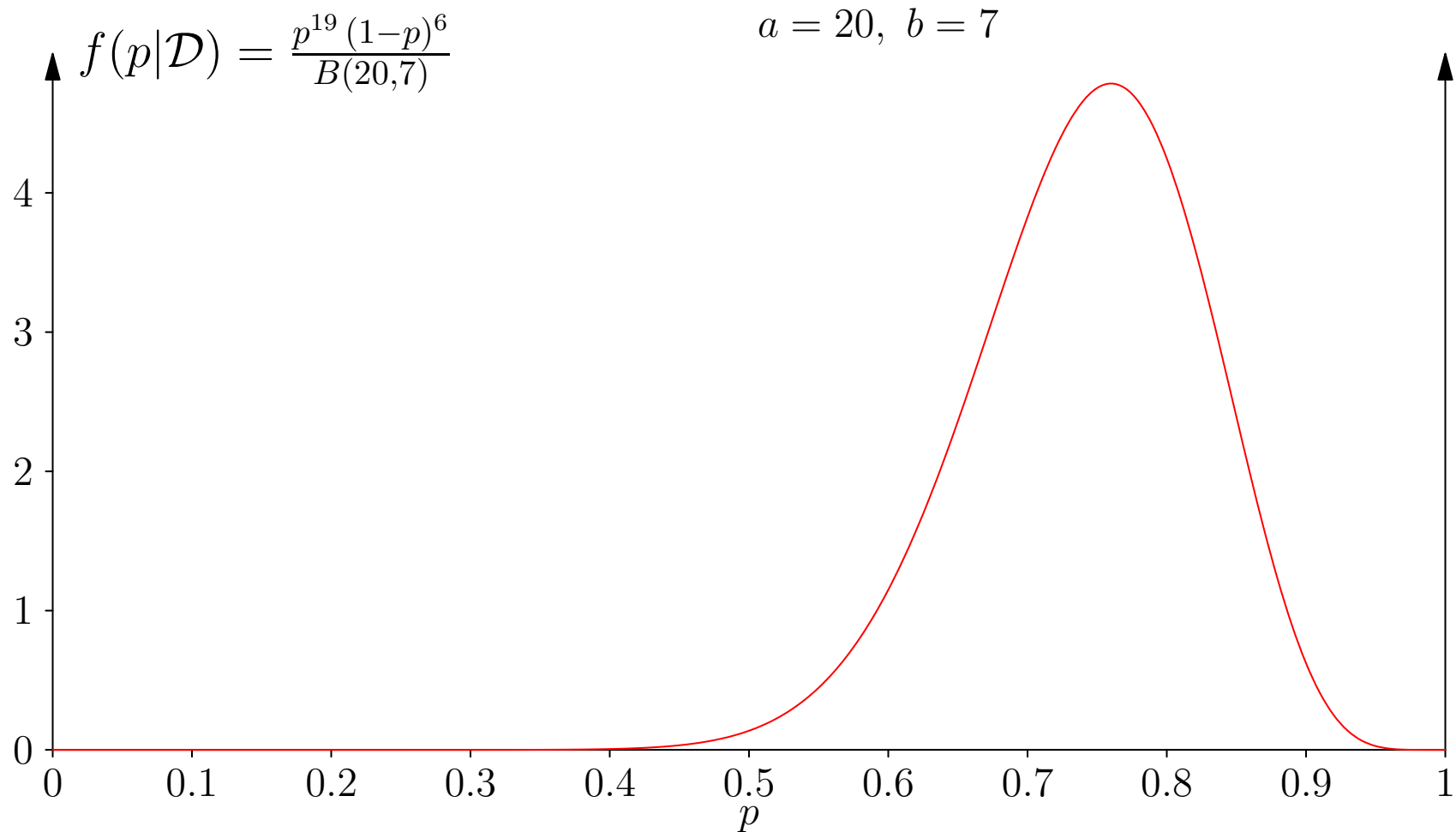
$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H, T, H, H, H\}$

$a = 19, b = 7$



Example ($p=0.7$)

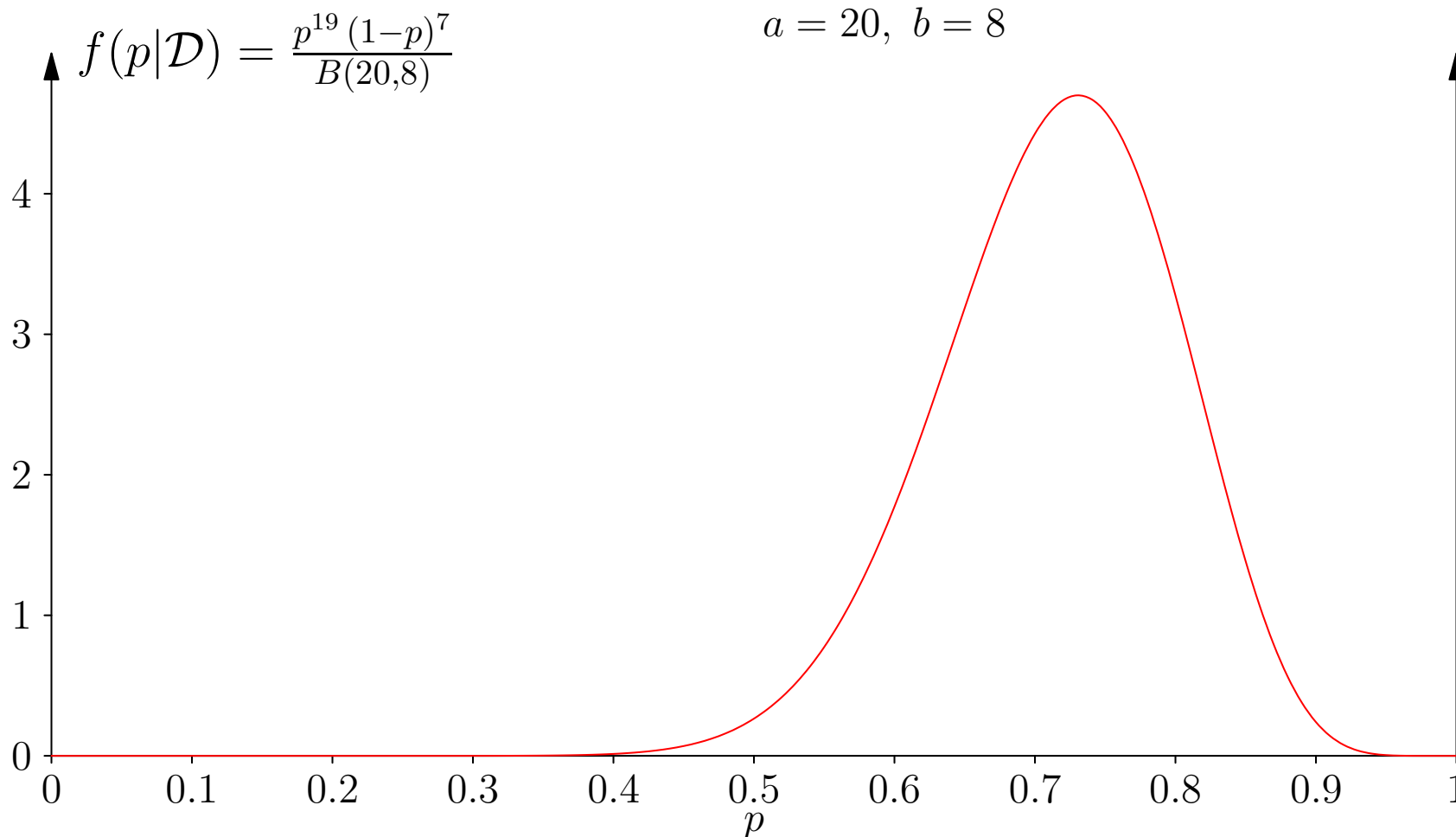
$\mathcal{D} = \{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,H,T,H,H,H,H,$



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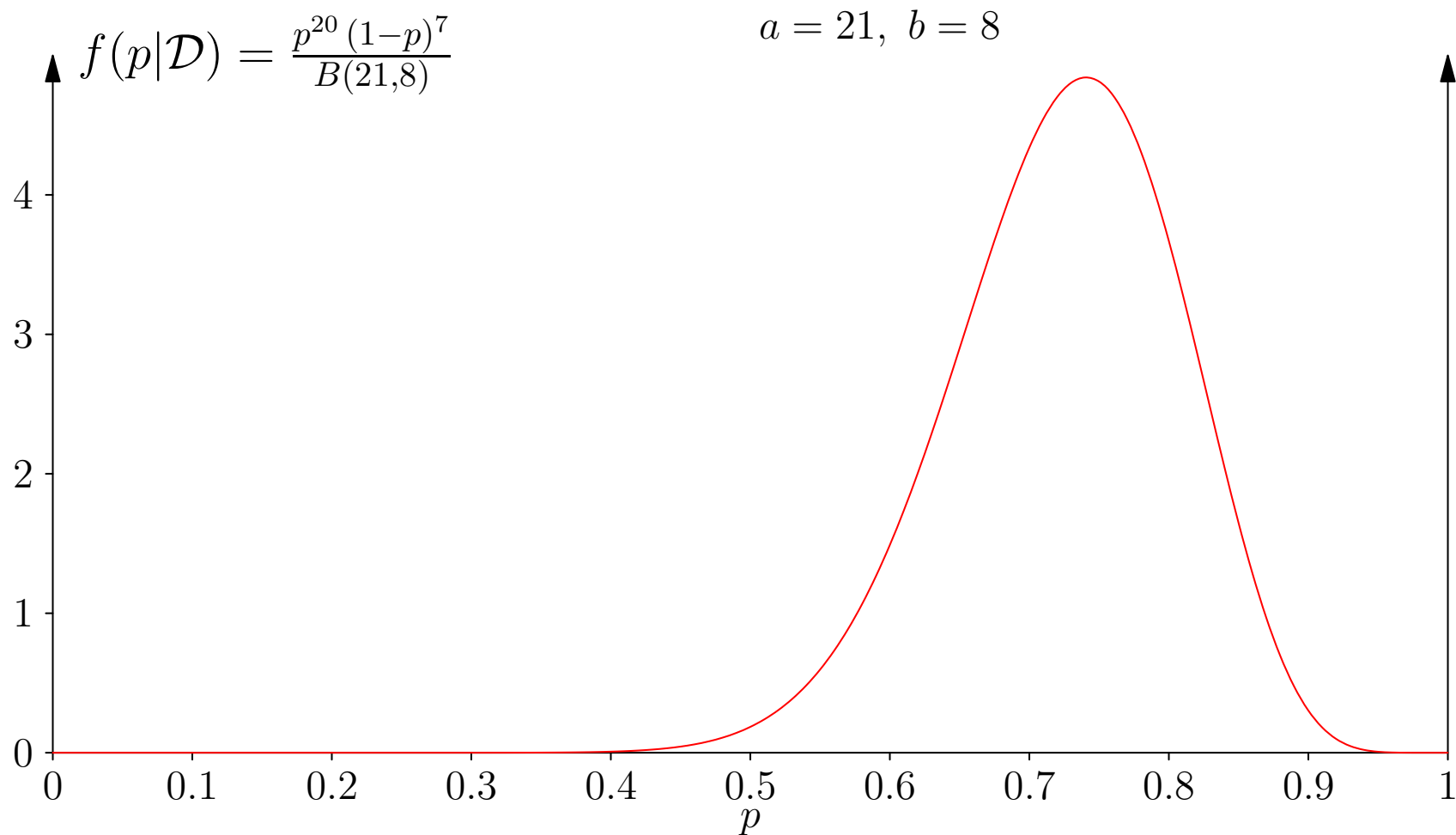
$\mathcal{D} = \{H, H, H, H, T, T, T, H, H, H, H, T, H, T, H, H, H, H, H, T, H, H, H, H, T\}$,

$a = 20, b = 8$



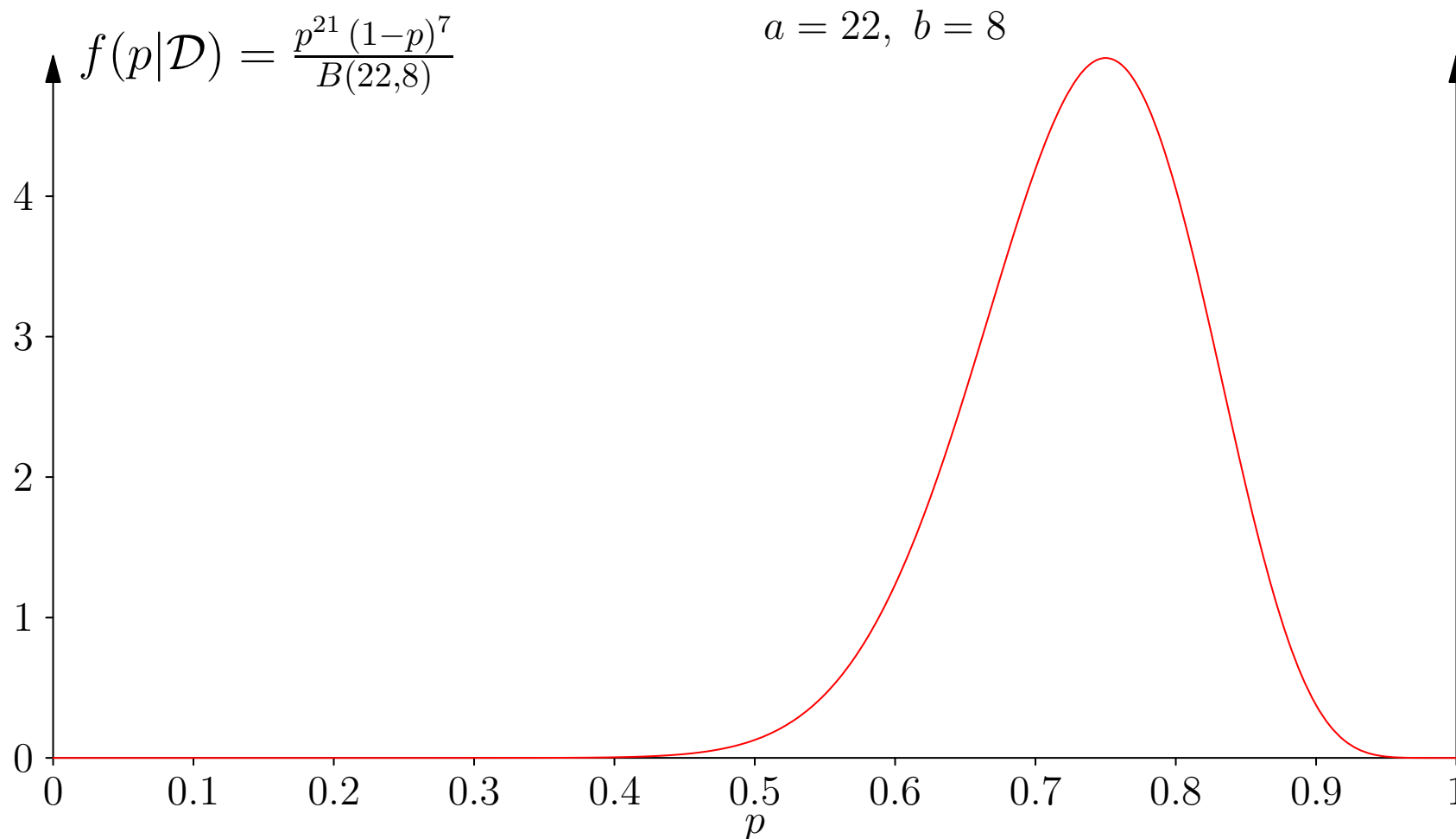
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$\mathcal{D} = \{H,H,H,H,T,T,T,H,H,H,H,T,H,T,H,H,H,H,H,T,H,H,H,H,T,H,$



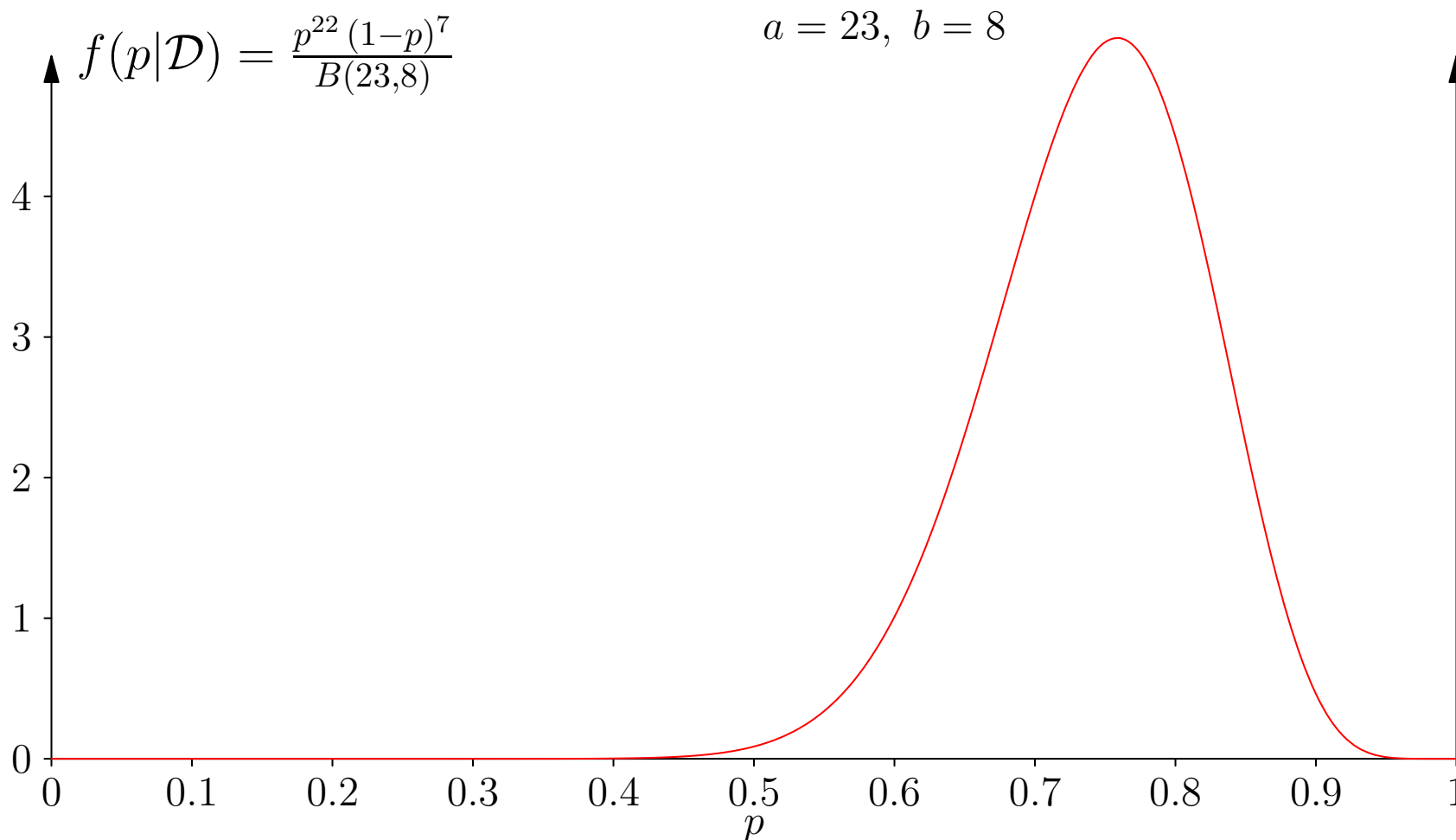
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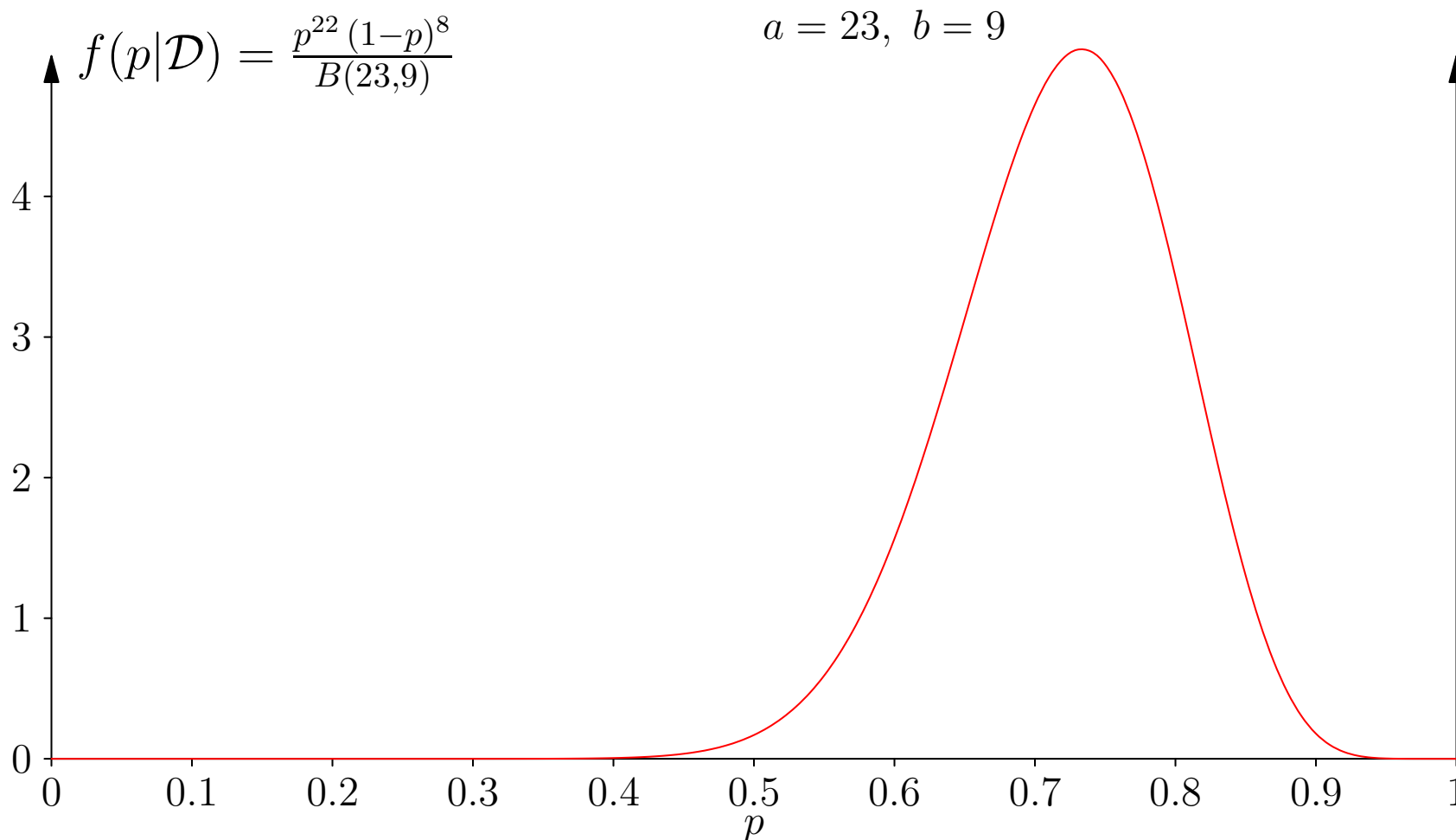
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Estimating Prediction Errors

- A full Bayesian treatment gives a prediction of its own error
- Assuming $f(p|\mathcal{D}) = \text{Beta}(p|a,b)$
- The expected value of p is given by $a/(a+b) = 23/32 = 0.72$
- The standard deviation is

$$\sqrt{\frac{ab}{(a+b)^2(a+b+1)}} = 0.08$$

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Poisson Likelihoods

- Let's look at a second example of conjugate priors
- Suppose we want to find the rate of traffic along a road between 1:00pm and 2:00pm
- We assume the number of cars is given by a Poisson distribution

$$\mathbb{P}(N) = \text{Pois}(N|\mu) = \frac{\mu^N}{N!}e^{-\mu}$$

- μ is the rate of traffic per hour which we want to infer from observation taken on different days

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Using Bayes

- Let us assume a Gamma distributed prior

$$p(\mu) = \Gamma(\mu|a_0, b_0) = \frac{b_0^{a_0} \mu^{a_0-1} e^{-b_0 \mu}}{\Gamma(a)}$$

- We will assume that we know nothing. The uninformative prior is $a_0 = b_0 = 0$
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Posterior

- The posterior after seeing the first piece of data is

$$\begin{aligned} p(\mu|N_1) &\propto \mathbb{P}(N_1|\mu)p(\mu) \\ &\propto \frac{\mu^{N_1}}{N_1!} e^{-\mu} \mu^{a_0-1} e^{-b_0\mu} \\ &\propto \mu^{N_1+a_0-1} e^{-(b_0+1)\mu} \end{aligned}$$

- The posterior is also a Gamma distribution $\Gamma(\mu|a_1, b_1)$ with $a_1 = a_0 + N_1$, $b_1 = b_0 + 1$

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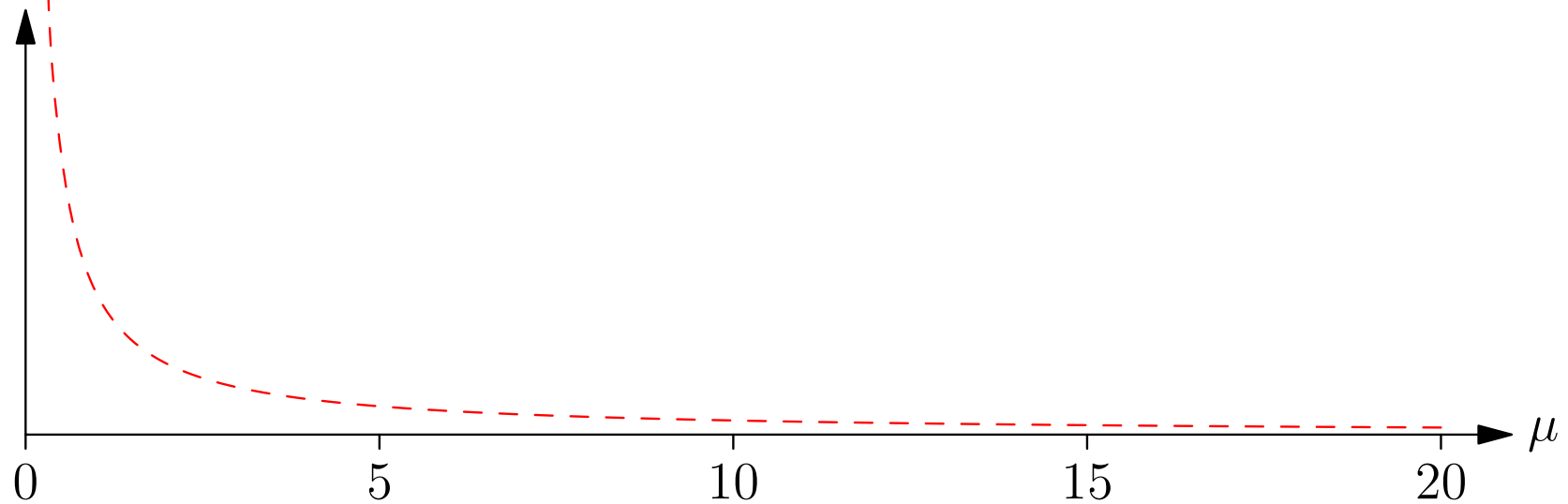
$$\begin{aligned} p(\mu|N_1) &\propto \mathbb{P}(N_1|\mu)p(\mu) \\ &\propto \frac{\mu^{N_1}}{N_1!} e^{-\mu} \mu^{a_0-1} e^{-b_0\mu} \\ &\propto \mu^{N_1+a_0-1} e^{-(b_0+1)\mu} \end{aligned}$$

- The posterior is also a Gamma distribution $\Gamma(\mu|a_1, b_1)$ with $a_1 = a_0 + N_1$, $b_1 = b_0 + 1$

Example ($\mu = 5$)

$$p(\mu|0, 0) = \frac{0^0 \mu^{-1} e^{-0 \mu}}{\Gamma(0)}$$

$$a = 0, \quad b = 0$$

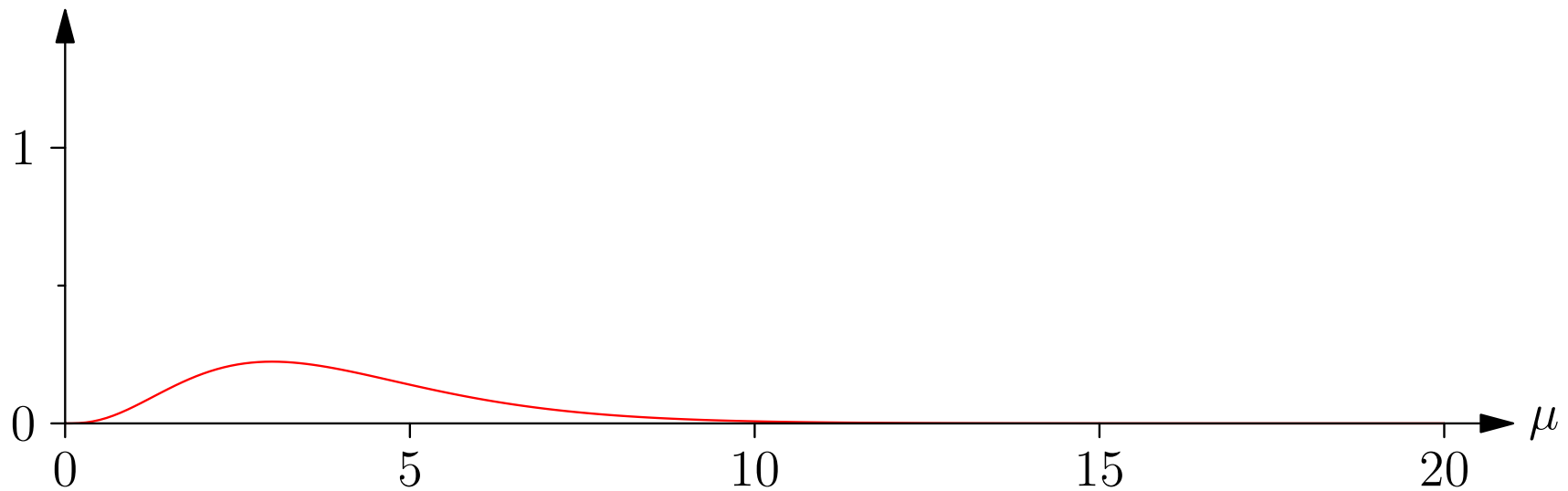


Example ($\mu = 5$)

$$\mathcal{D} = \{4\}$$

$$p(\mu|4, 1) = \frac{1^4 \mu^3 e^{-1\mu}}{\Gamma(4)}$$

$$a = 4, \quad b = 1$$

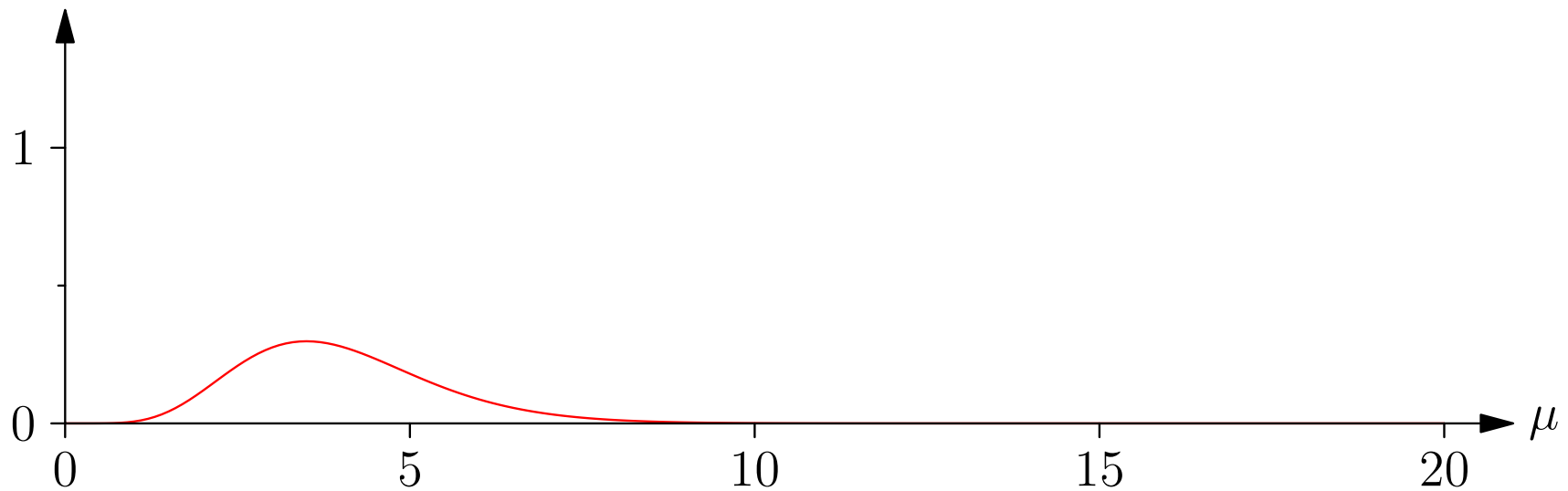


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4\}$$

$$p(\mu|8, 2) = \frac{2^8 \mu^7 e^{-2\mu}}{\Gamma(8)}$$

$$a = 8, \quad b = 2$$

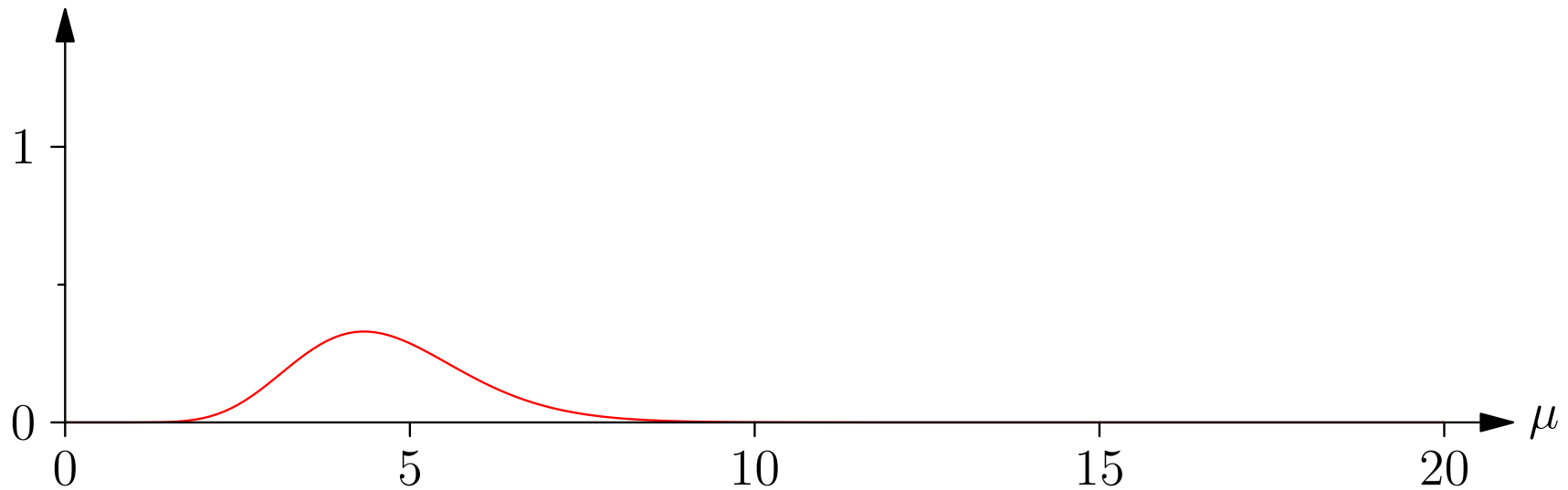


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6\}$$

$$p(\mu|14, 3) = \frac{3^{14} \mu^{13} e^{-3\mu}}{\Gamma(14)}$$

$$a = 14, \quad b = 3$$

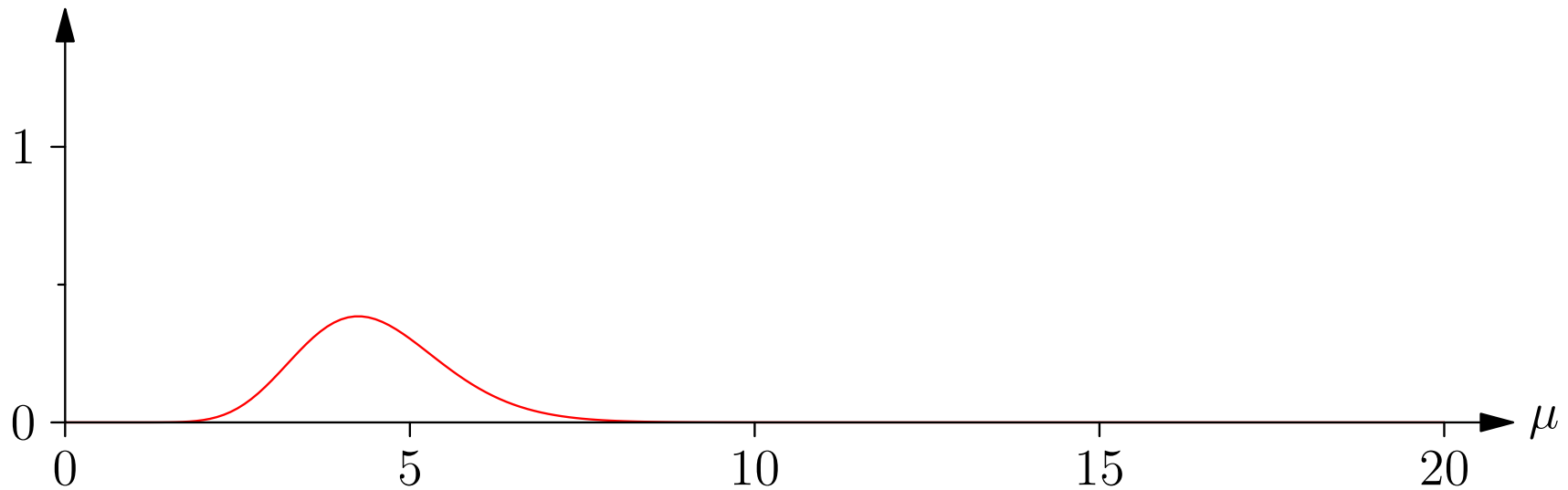


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4\}$$

$$p(\mu|18, 4) = \frac{4^{18} \mu^{17} e^{-4\mu}}{\Gamma(18)}$$

$$a = 18, \quad b = 4$$

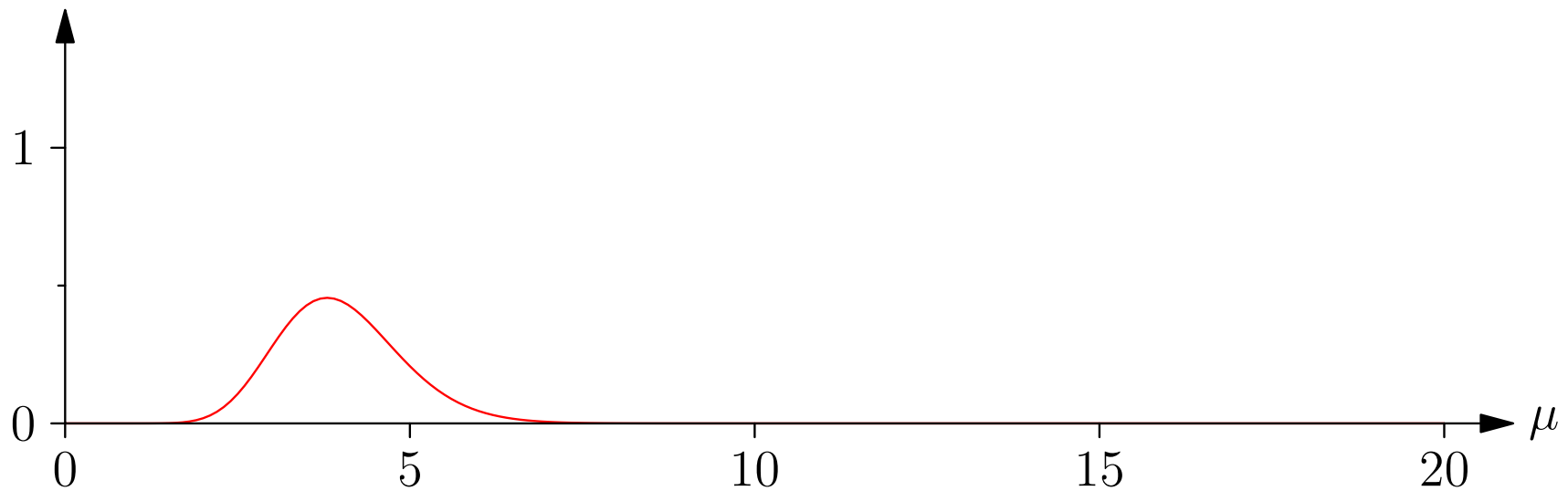


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2\}$$

$$p(\mu|20, 5) = \frac{5^{20} \mu^{19} e^{-5 \mu}}{\Gamma(20)}$$

$$a = 20, \quad b = 5$$

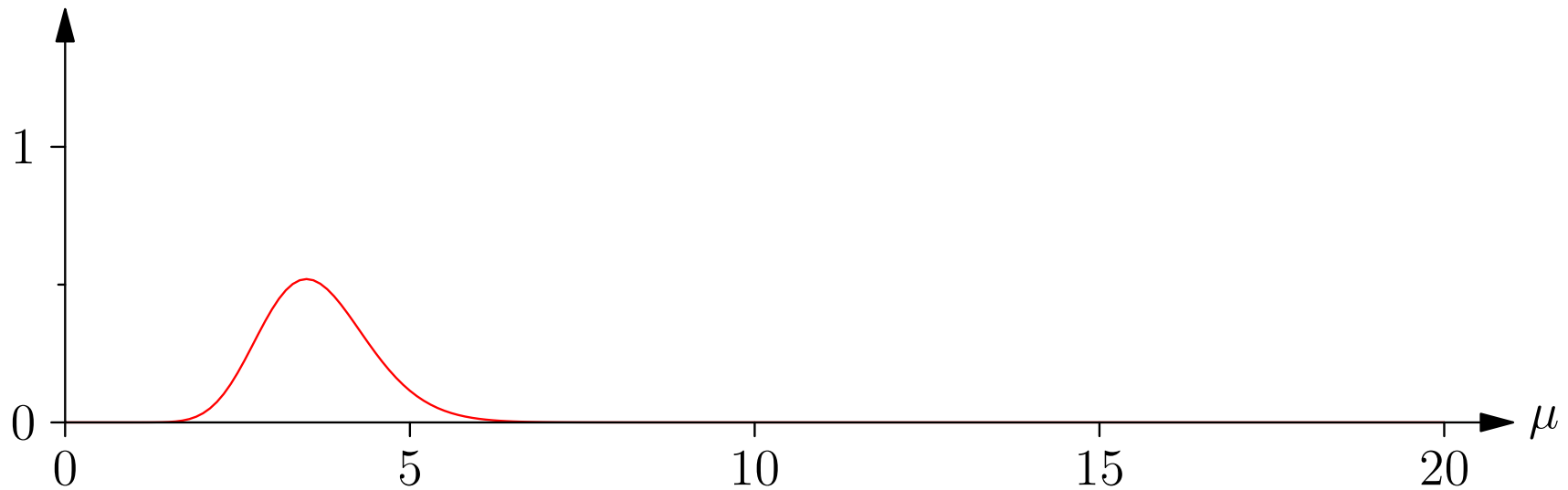


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2\}$$

$$p(\mu|22, 6) = \frac{6^{22} \mu^{21} e^{-6\mu}}{\Gamma(22)}$$

$$a = 22, \quad b = 6$$

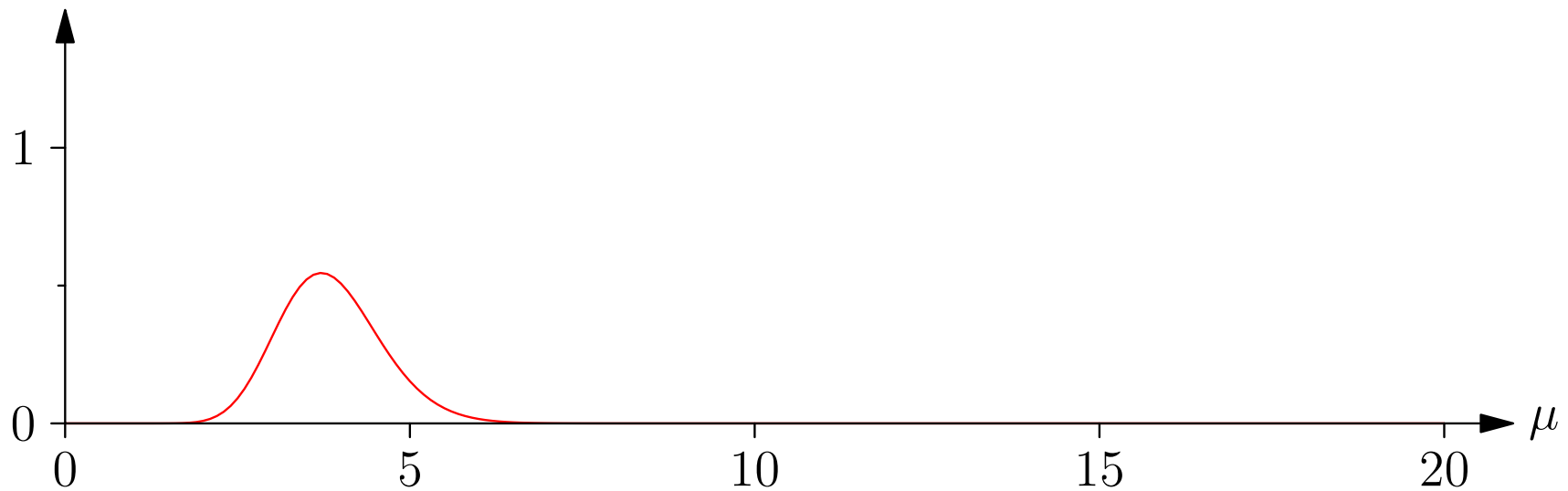


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5\}$$

$$p(\mu|27, 7) = \frac{7^{27} \mu^{26} e^{-7 \mu}}{\Gamma(27)}$$

$$a = 27, \quad b = 7$$

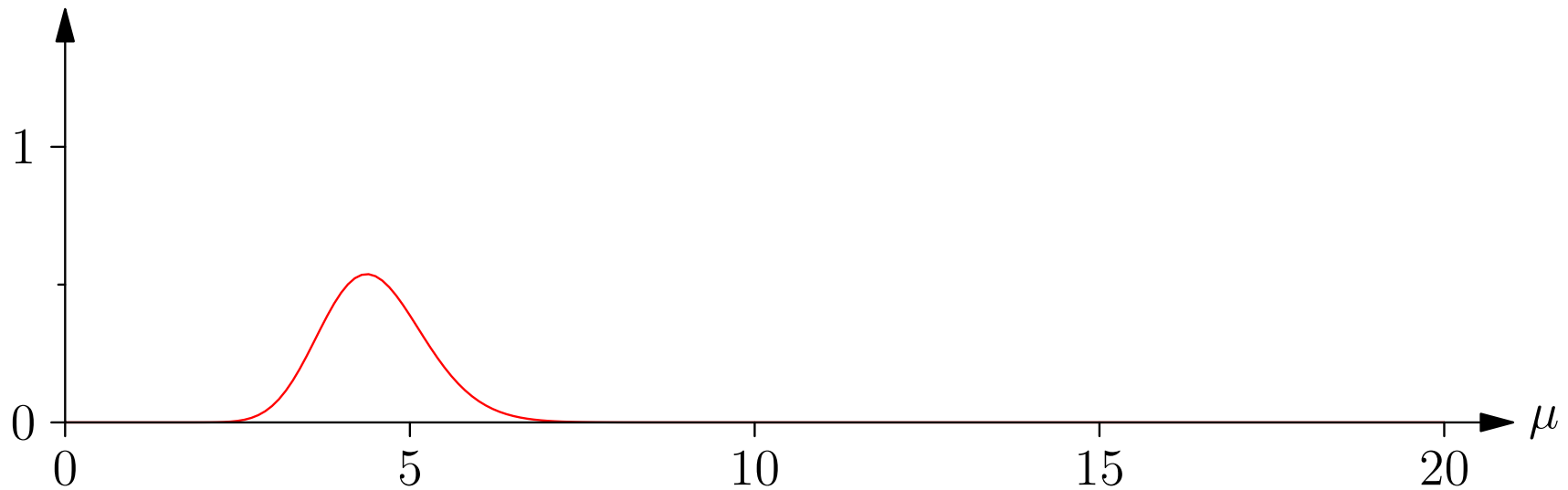


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9\}$$

$$p(\mu|36, 8) = \frac{8^{36} \mu^{35} e^{-8\mu}}{\Gamma(36)}$$

$$a = 36, \quad b = 8$$

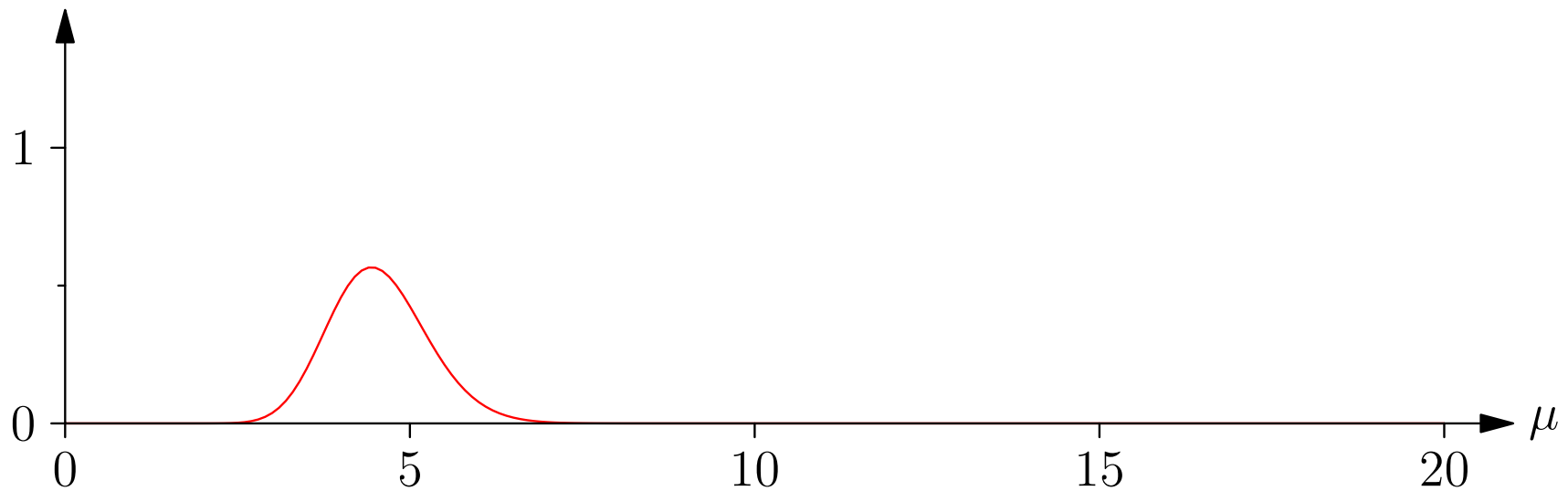


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5\}$$

$$p(\mu|41, 9) = \frac{9^{41} \mu^{40} e^{-9\mu}}{\Gamma(41)}$$

$$a = 41, \quad b = 9$$

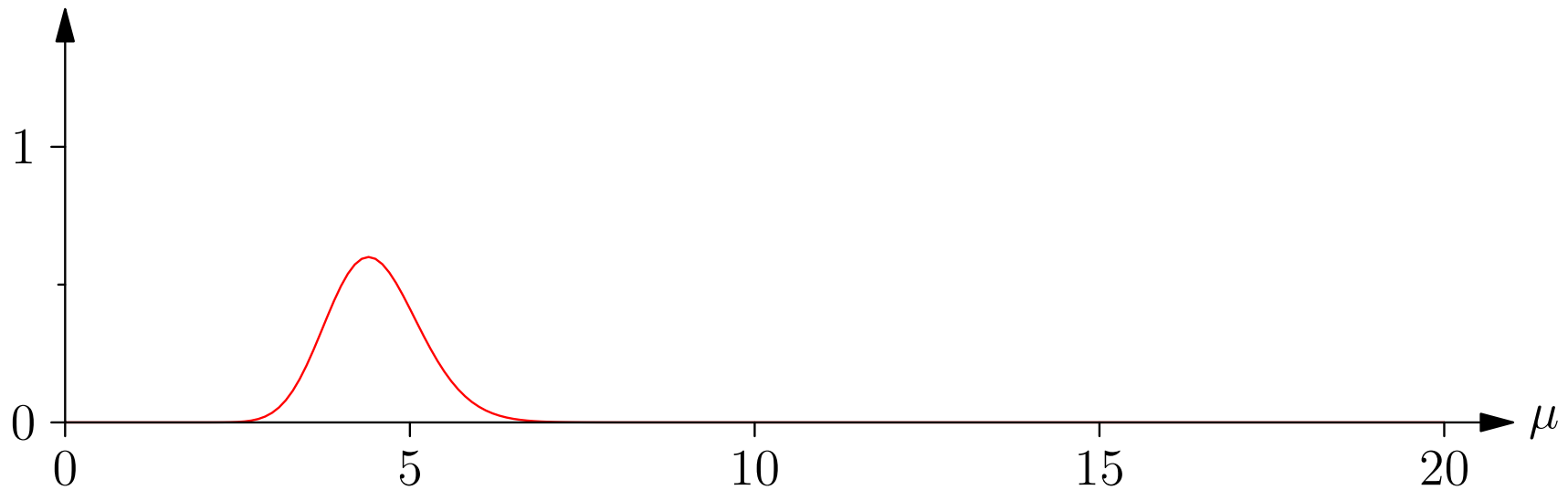


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4\}$$

$$p(\mu|45, 10) = \frac{10^{45} \mu^{44} e^{-10 \mu}}{\Gamma(45)}$$

$$a = 45, \quad b = 10$$

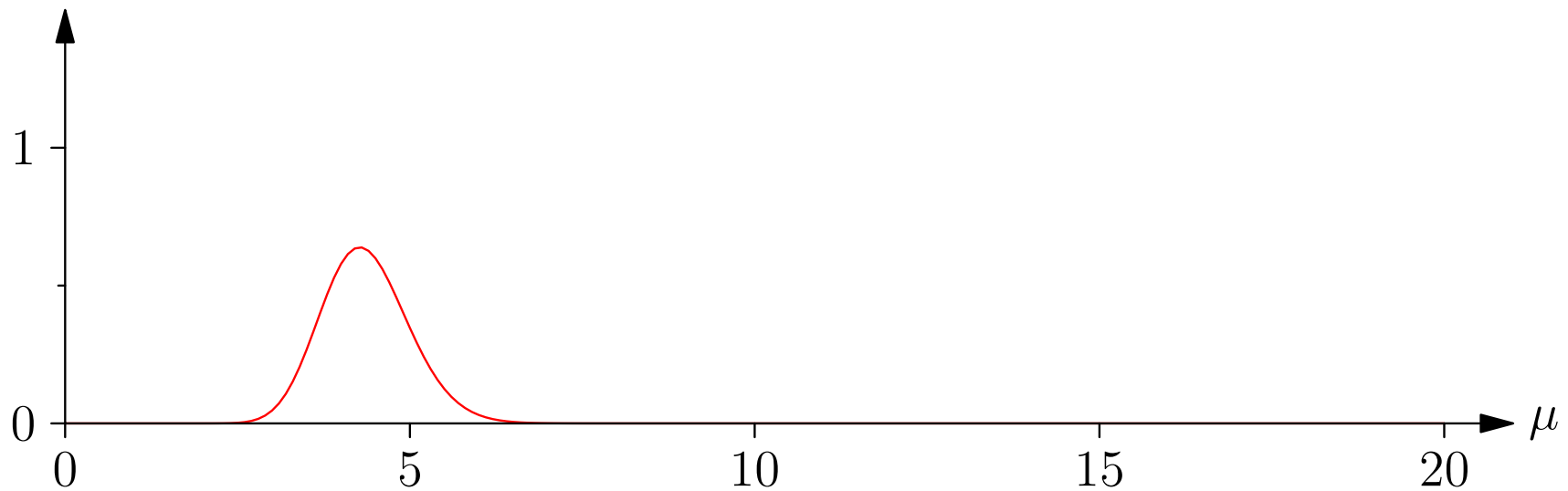


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3\}$$

$$p(\mu|48, 11) = \frac{11^{48} \mu^{47} e^{-11\mu}}{\Gamma(48)}$$

$$a = 48, \quad b = 11$$

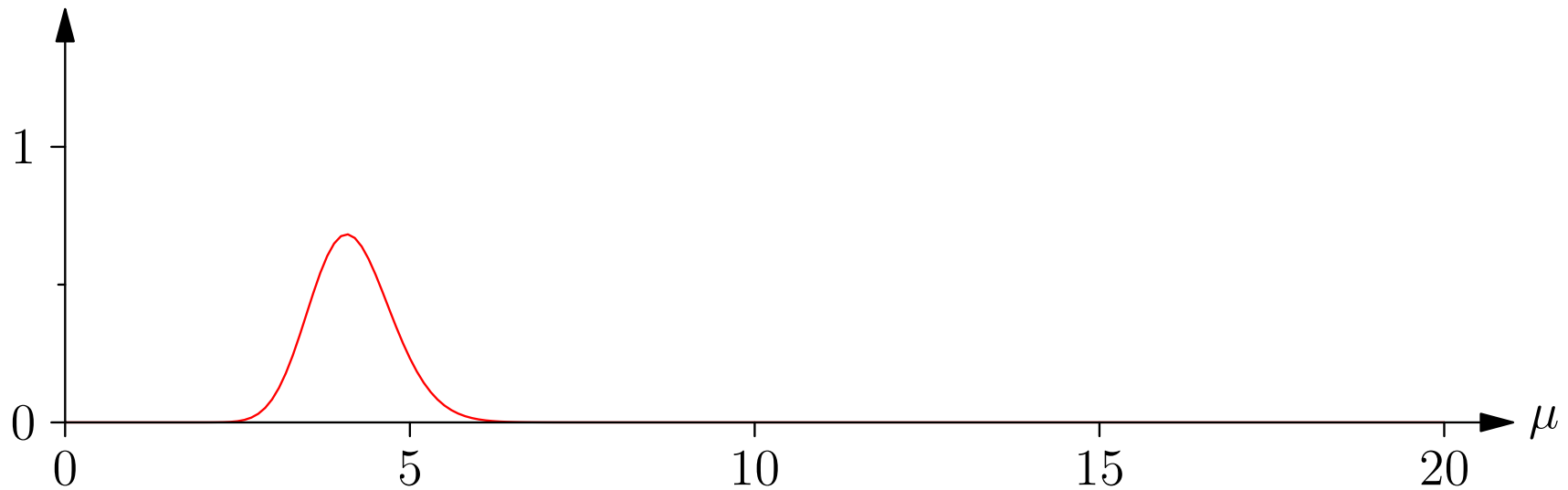


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2\}$$

$$p(\mu|50, 12) = \frac{12^{50} \mu^{49} e^{-12 \mu}}{\Gamma(50)}$$

$$a = 50, \quad b = 12$$

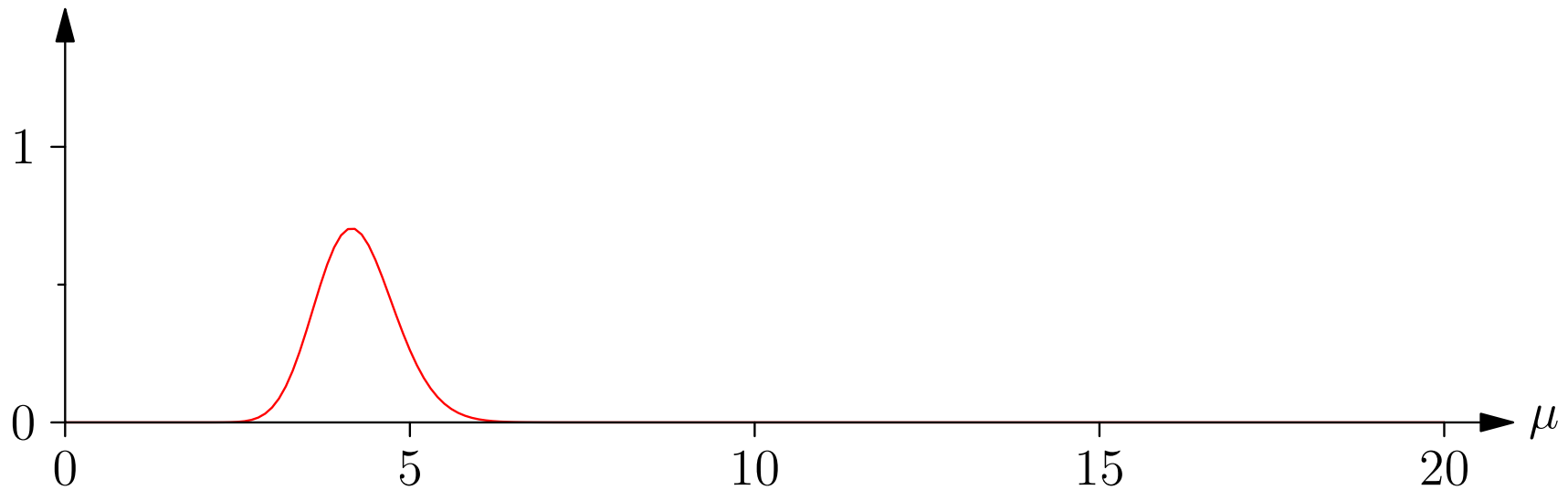


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5\}$$

$$p(\mu|55, 13) = \frac{13^{55} \mu^{54} e^{-13\mu}}{\Gamma(55)}$$

$$a = 55, \quad b = 13$$

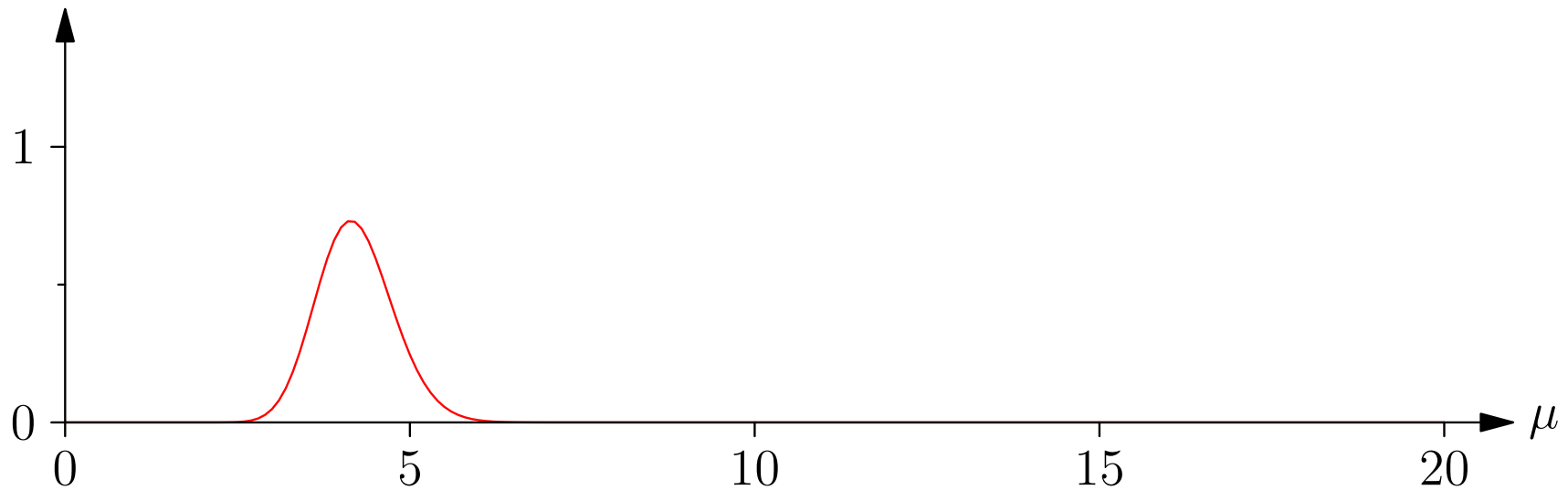


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4\}$$

$$p(\mu|59, 14) = \frac{14^{59} \mu^{58} e^{-14\mu}}{\Gamma(59)}$$

$$a = 59, \quad b = 14$$

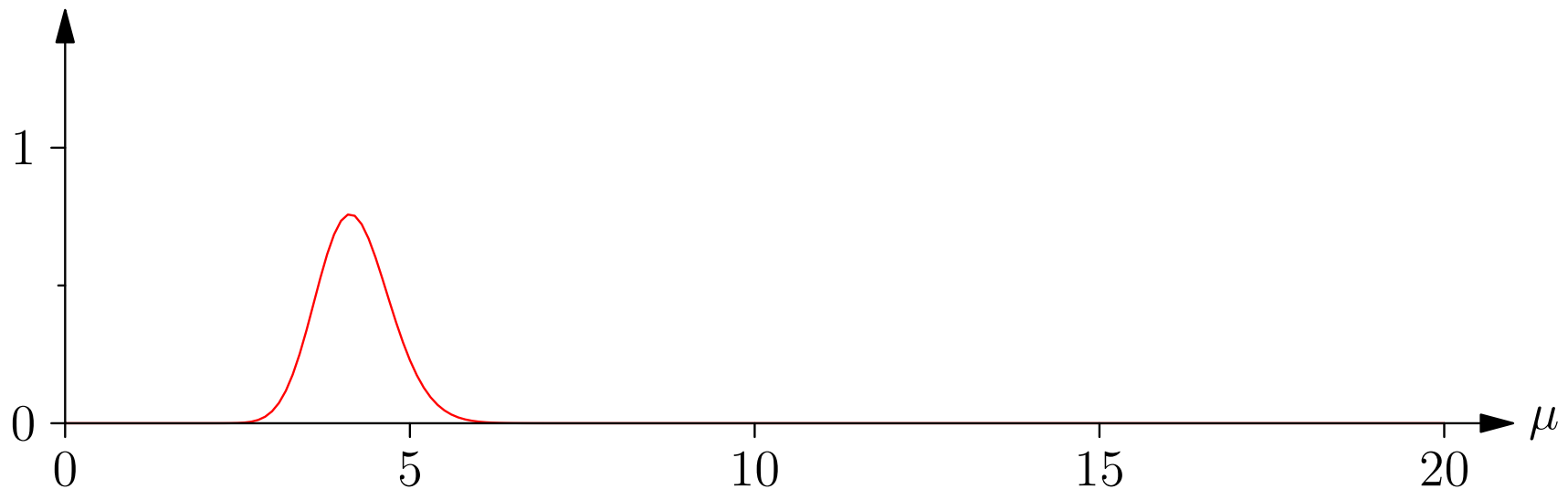


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4\}$$

$$p(\mu|63, 15) = \frac{15^{63} \mu^{62} e^{-15\mu}}{\Gamma(63)}$$

$$a = 63, \quad b = 15$$

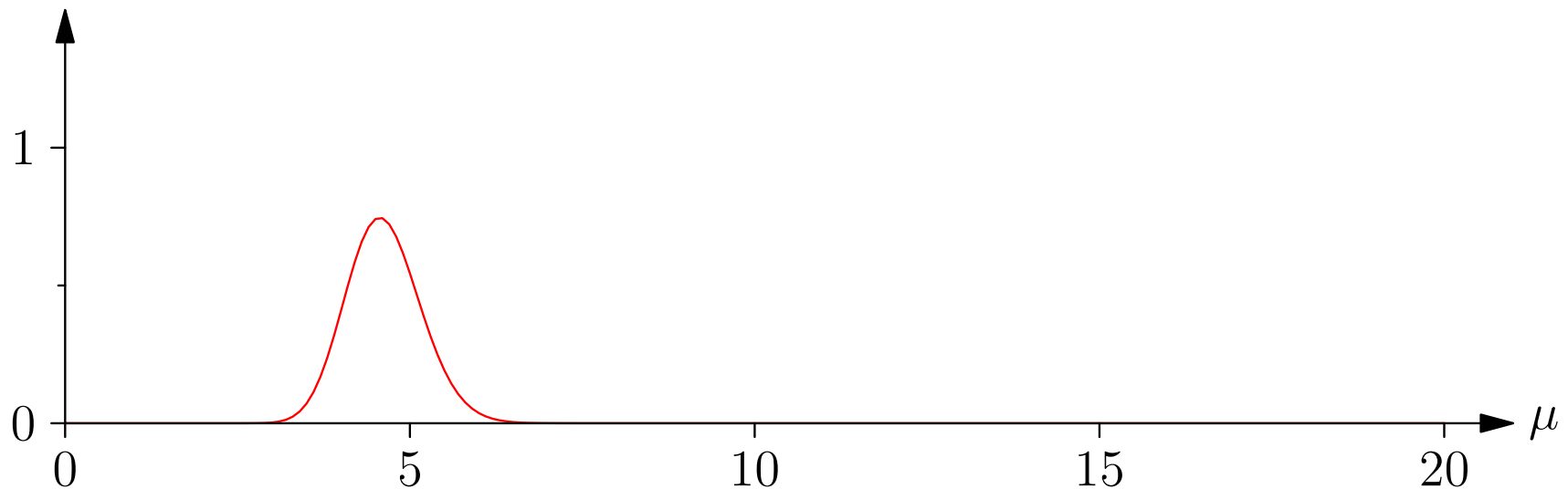


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11\}$$

$$p(\mu|74, 16) = \frac{16^{74} \mu^{73} e^{-16\mu}}{\Gamma(74)}$$

$$a = 74, \quad b = 16$$

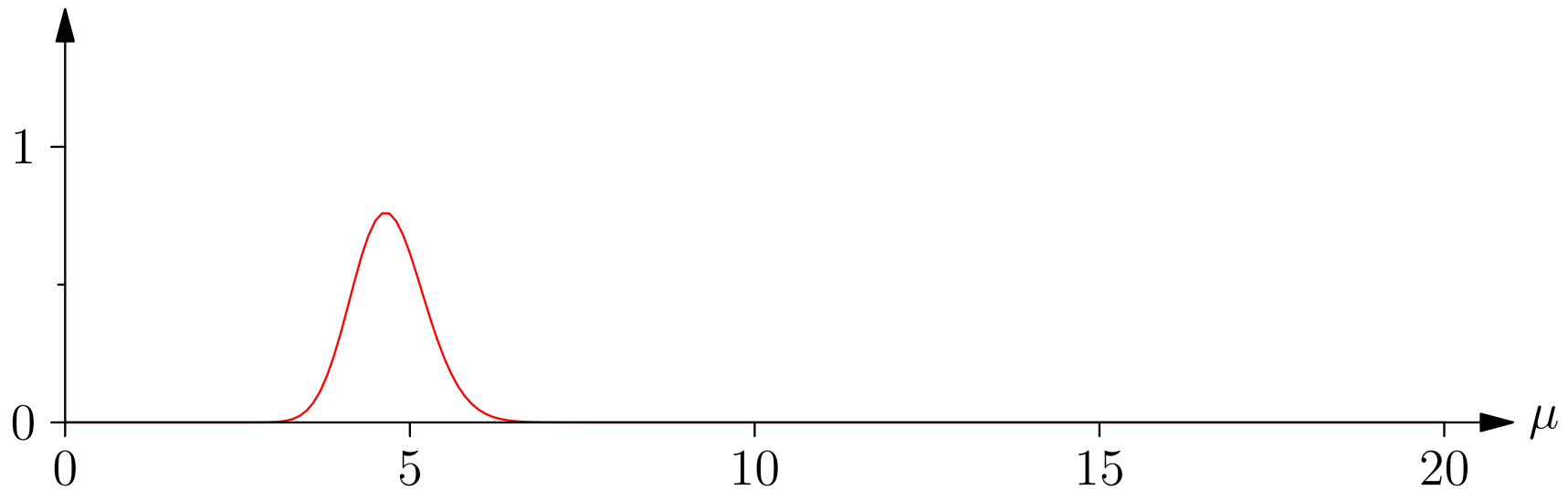


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6\}$$

$$p(\mu|80, 17) = \frac{17^{80} \mu^{79} e^{-17 \mu}}{\Gamma(80)}$$

$$a = 80, \quad b = 17$$

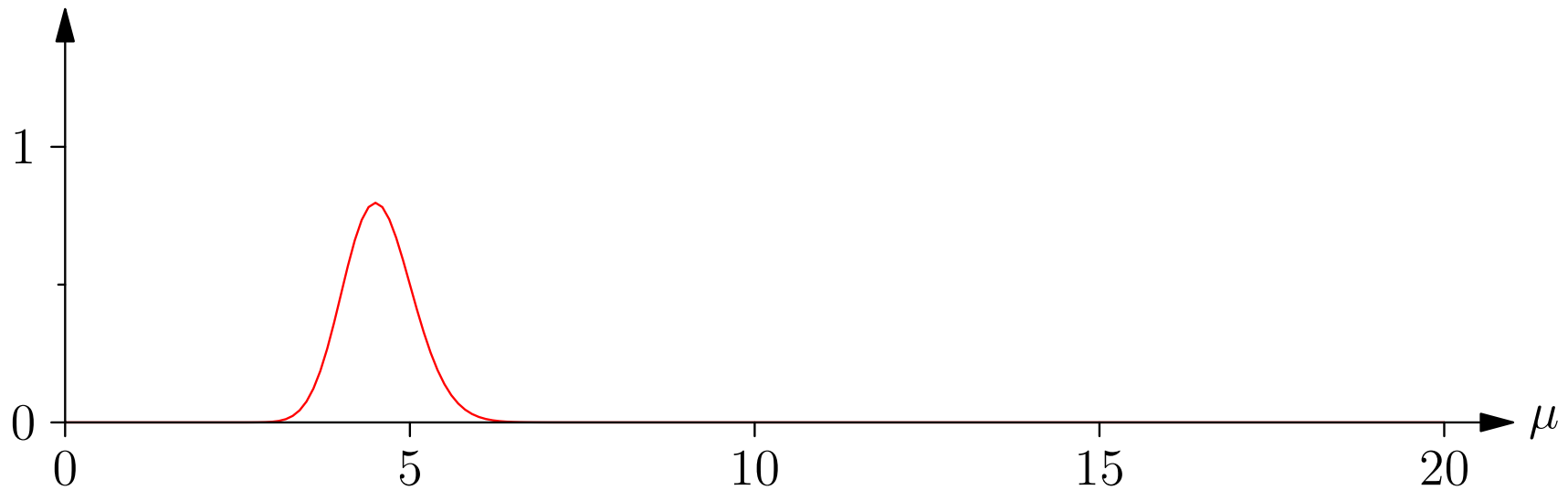


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2\}$$

$$p(\mu|82, 18) = \frac{18^{82} \mu^{81} e^{-18\mu}}{\Gamma(82)}$$

$$a = 82, \quad b = 18$$

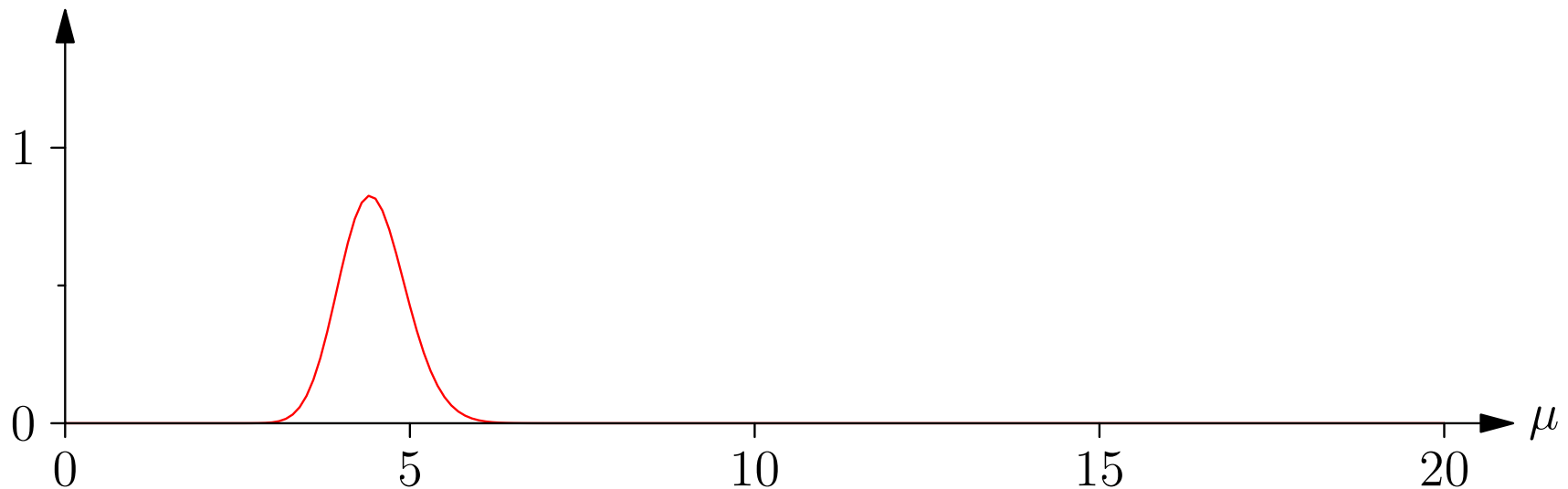


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3\}$$

$$p(\mu|85, 19) = \frac{19^{85} \mu^{84} e^{-19 \mu}}{\Gamma(85)}$$

$$a = 85, \quad b = 19$$

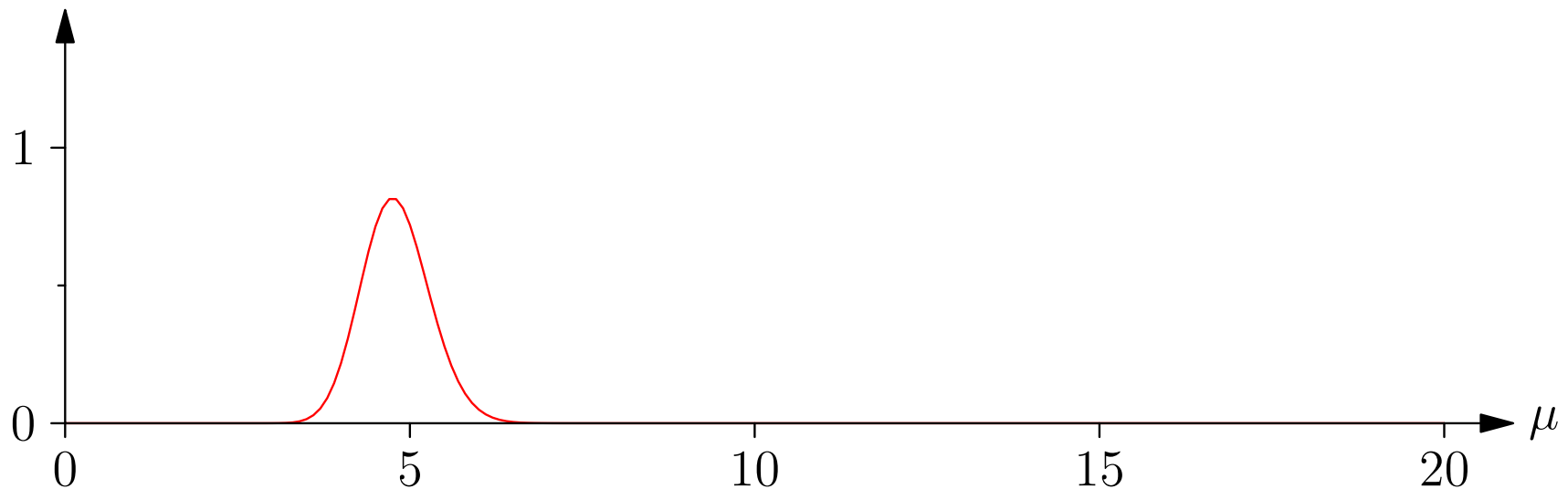


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$

$$p(\mu|96, 20) = \frac{20^{96} \mu^{95} e^{-20 \mu}}{\Gamma(96)}$$

$$a = 96, \quad b = 20$$

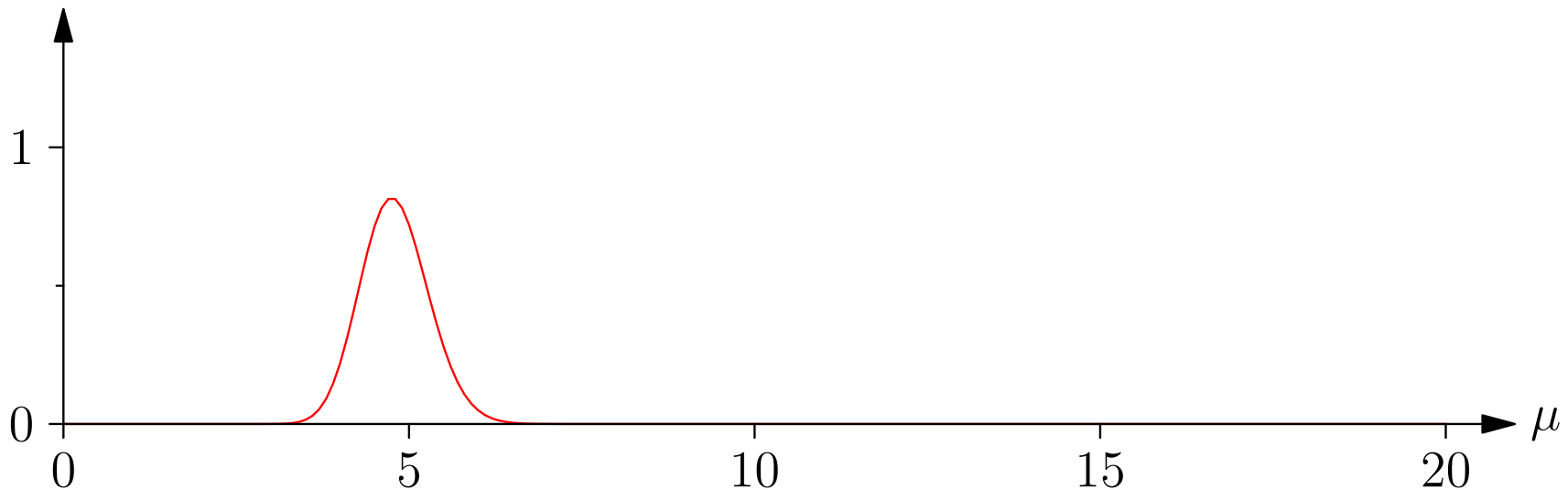


Example ($\mu = 5$)

$$\mathcal{D} = \{4, 4, 6, 4, 2, 2, 5, 9, 5, 4, 3, 2, 5, 4, 4, 11, 6, 2, 3, 11\}$$

$$p(\mu|96, 20) = \frac{20^{96} \mu^{95} e^{-20 \mu}}{\Gamma(96)}$$

$$a = 96, \quad b = 20$$



$$\mathbb{E}[\mu] = \frac{a}{b} = \frac{96}{20} = 4.8$$

$$\sqrt{\text{Var}(\mu)} = \sqrt{\frac{a}{b^2}} = 0.5$$

Outline

1. Bayes' Rule
2. Conjugate Priors
3. **Uninformative Priors**



Uninformative Priors

- What if we have no prior knowledge, what should we do?
- OK usually we know whether we should make a measurement using a micrometer, ruler or car mileage, but we might still know almost nothing
- This led to Bayesian statistics being labelled as *subjective*
- However Ed. Jaynes (the greatest proponent of Bayesian methods) argued that we could answer this using symmetry arguments

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Uninformative Priors for Scale Parameter

- Why did we choose $a_0 = b_0 = 0$ implying a prior $p(\mu) = 1/\mu$?



- That is, we have no idea on what scale to measure μ

$$\int_A^B p(\mu) d\mu = \int_{A/c}^{B/c} p(\mu) d\mu$$

- Or $p(\mu) = \frac{1}{c} p(\frac{\mu}{c})$ implying $p(\mu) \propto \frac{1}{\mu}$

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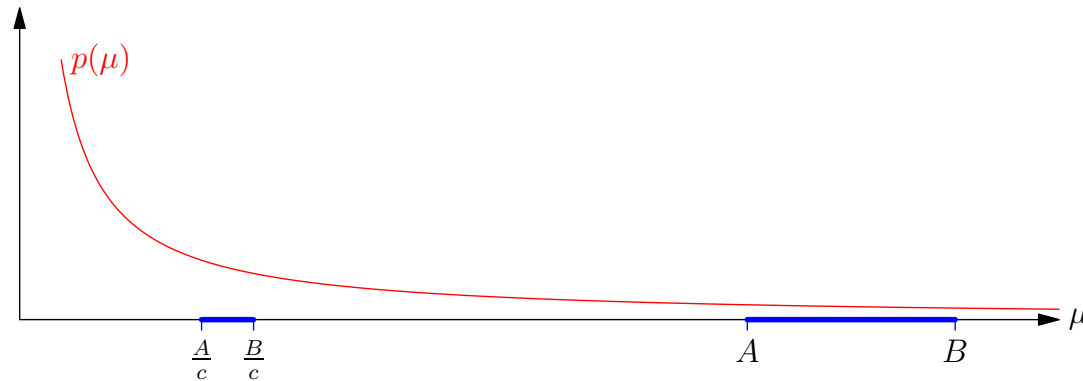
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- Or $p(\mu) = \frac{1}{c} p(\frac{\mu}{c})$ implying $p(\mu) \propto \frac{1}{\mu}$

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- Why did we choose $a_0 = b_0 = 0$ implying a prior $p(\mu) = 1/\mu$?



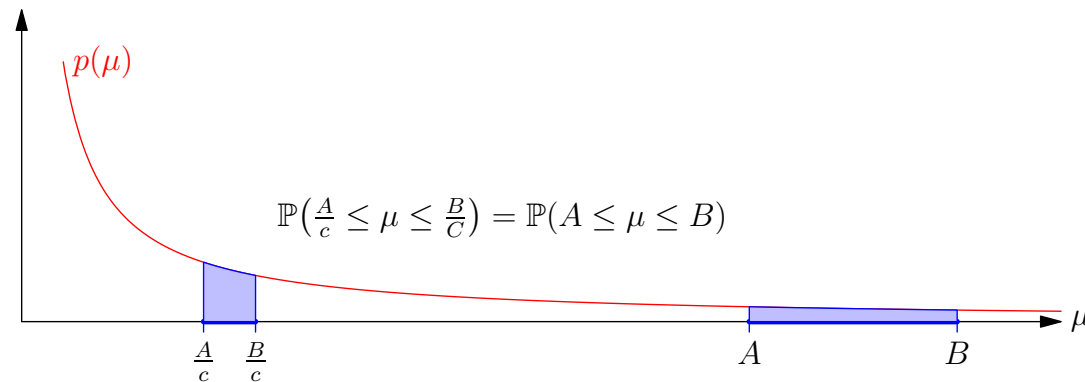
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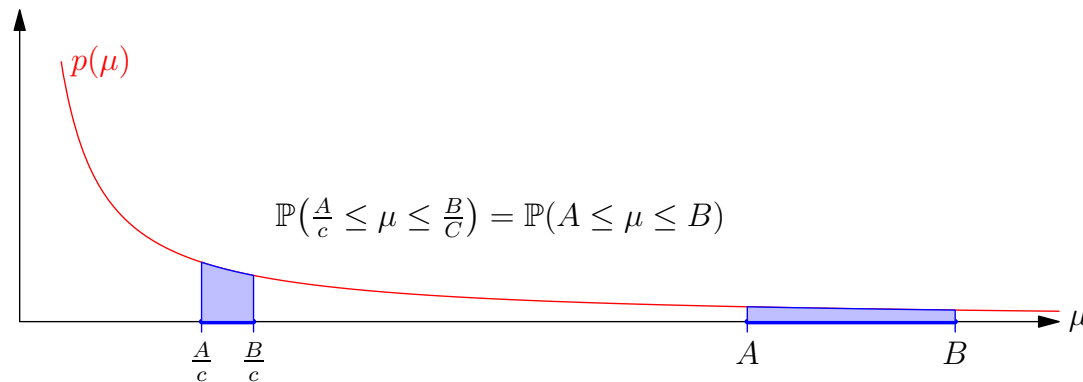
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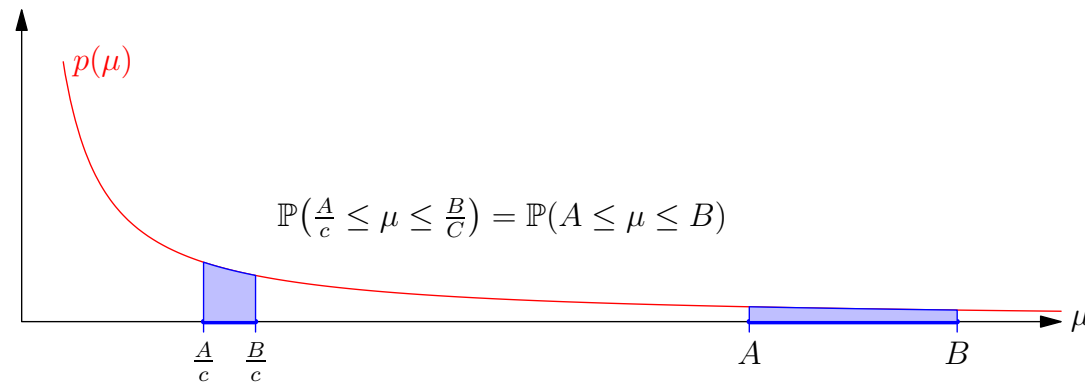
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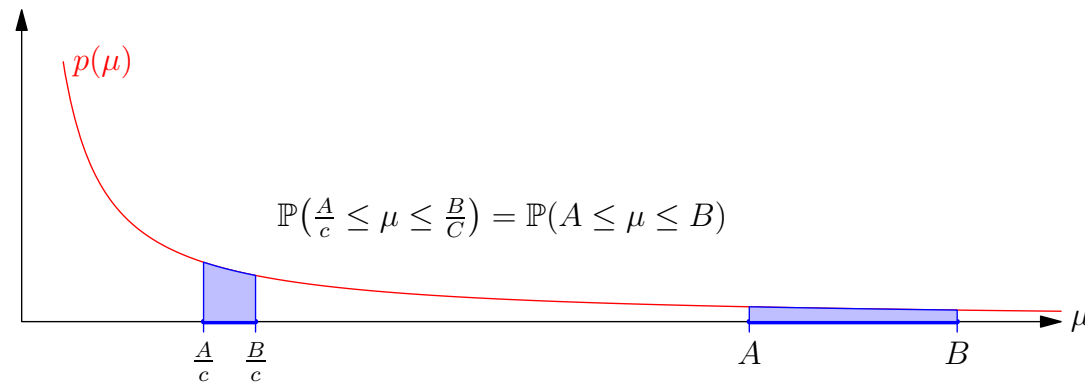
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Benford's Law

- Numbers occurring in life (physical constants, amounts of money) should not depend on the units (scale) measuring them
- They should then be distributed as $p(x) \propto 1/x$
- A curious consequence of this is that the significant figure has a distribution

$$\begin{aligned}\mathbb{P}(\text{most s.f. of } x = n) &= \frac{\int_n^{n+1} \frac{1}{x} dx}{\int_1^{10} \frac{1}{x} dx} \\ &= \frac{\log(n+1) - \log(n)}{\log(10)} = \log_{10}\left(\frac{n+1}{n}\right)\end{aligned}$$

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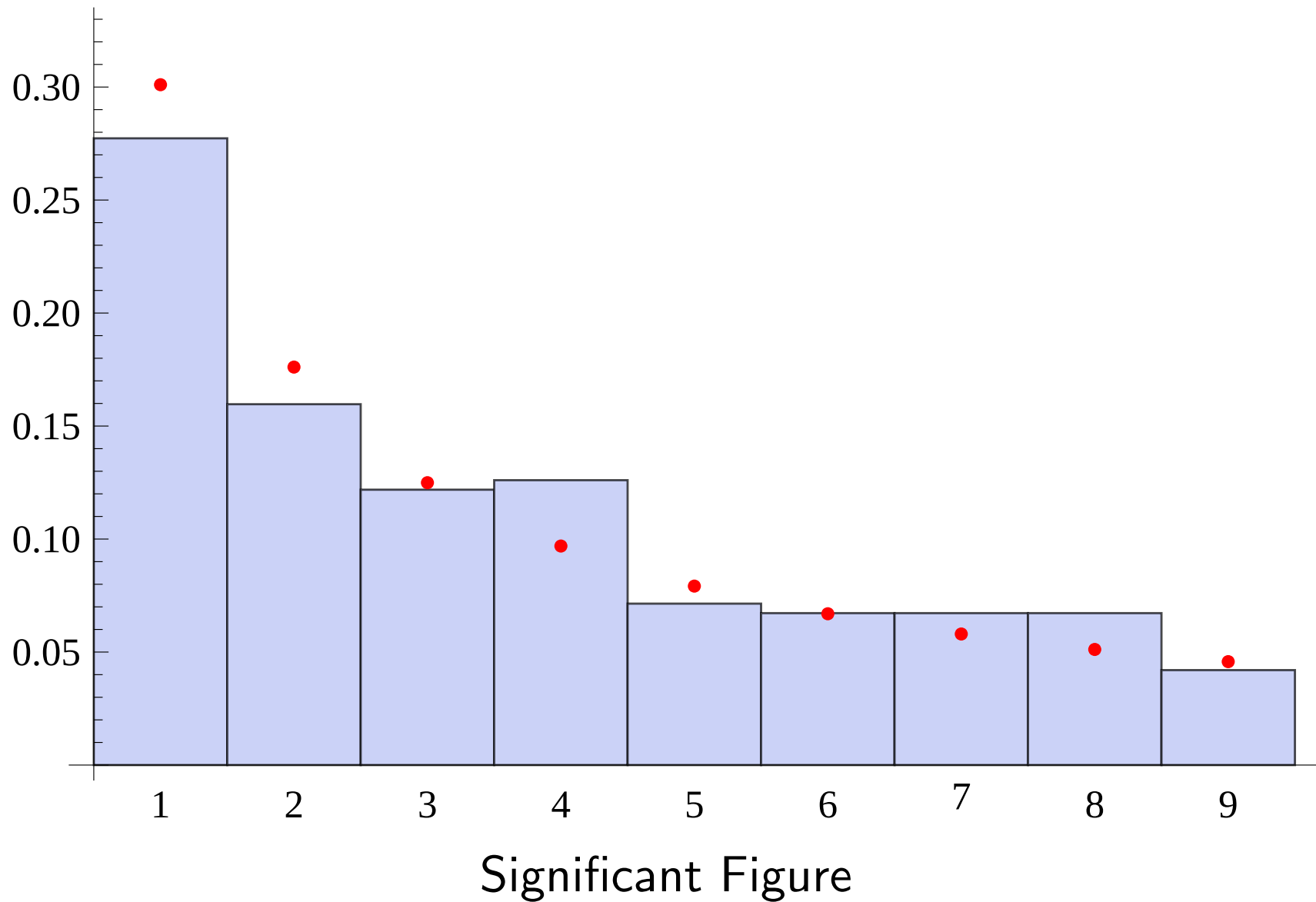
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Population Size of 238 Countries



Conclusion

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- However, it requires a model of what is happening
- In practice Bayesian methods are easy if the data is generated from a likelihood with a conjugate prior distribution—we have to be clever to choose the right prior
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- When we have no knowledge there are consistent ways to express our ignorance

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